

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-14 13:30:44

Data & configuration

LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	— (none)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	none
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	on · copies=2 · amplitude_scale, baseline_offset
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	Baseline, Shallow, Deep, BN_GAP, WideKernel, CNN_LSTM, CNN_BiLSTM
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

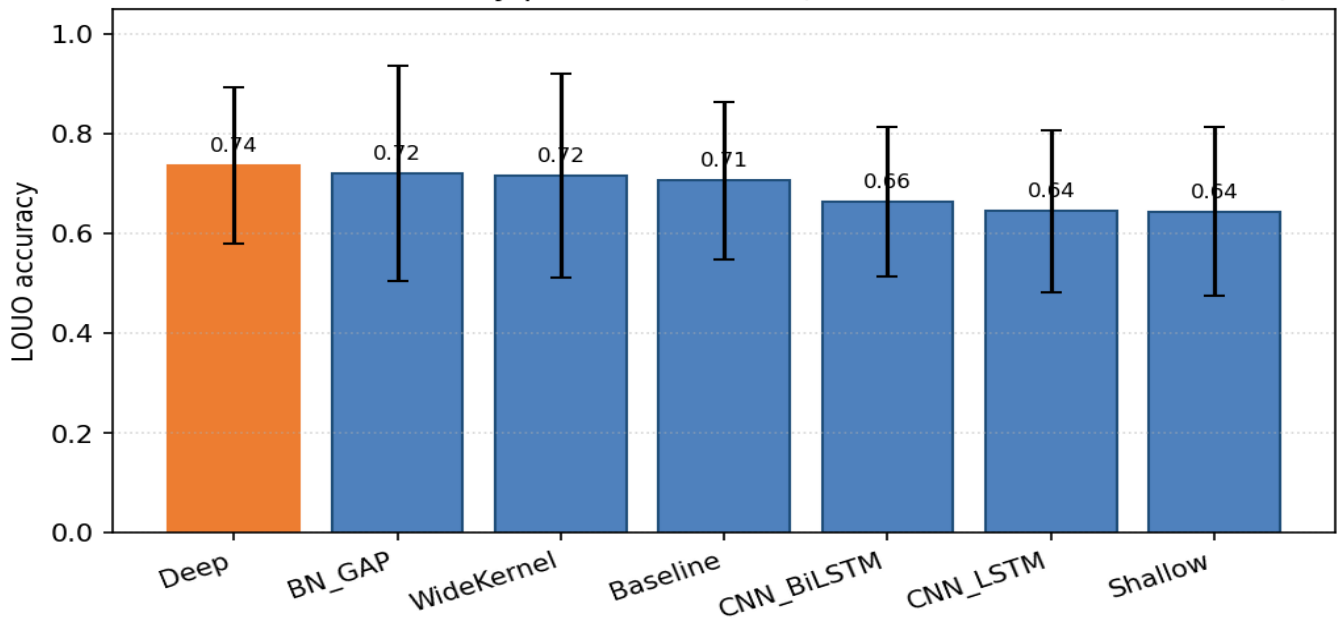
Variant comparison (LOUO)

Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
Deep	204,007	0.7355	0.1565	13	—
BN_GAP	57,319	0.7206	0.2161	13	—
WideKernel	131,687	0.7150	0.2035	13	—
Baseline	113,255	0.7050	0.1587	13	—
CNN_BiLSTM	105,575	0.6629	0.1500	13	—
CNN_LSTM	68,455	0.6441	0.1621	13	—
Shallow	72,487	0.6431	0.1689	13	—

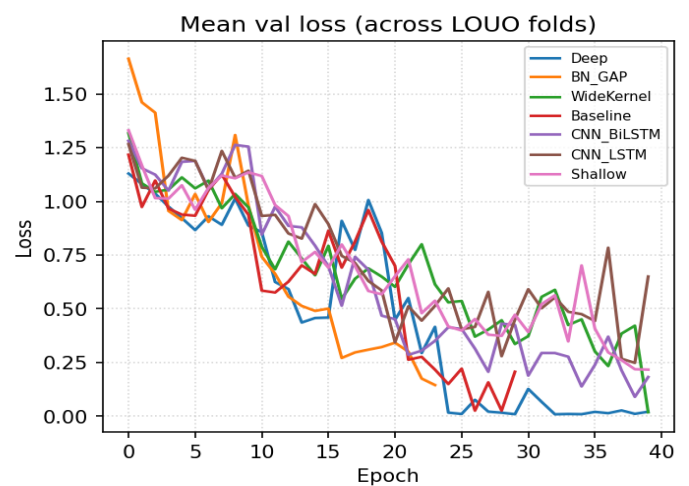
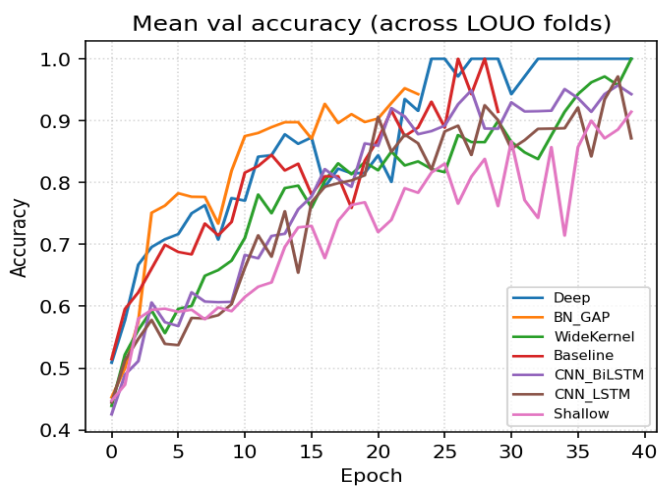
Best LOUO mean: Deep

LOUO accuracy per variant

LOUO mean accuracy per architecture (error bars = std across folds)

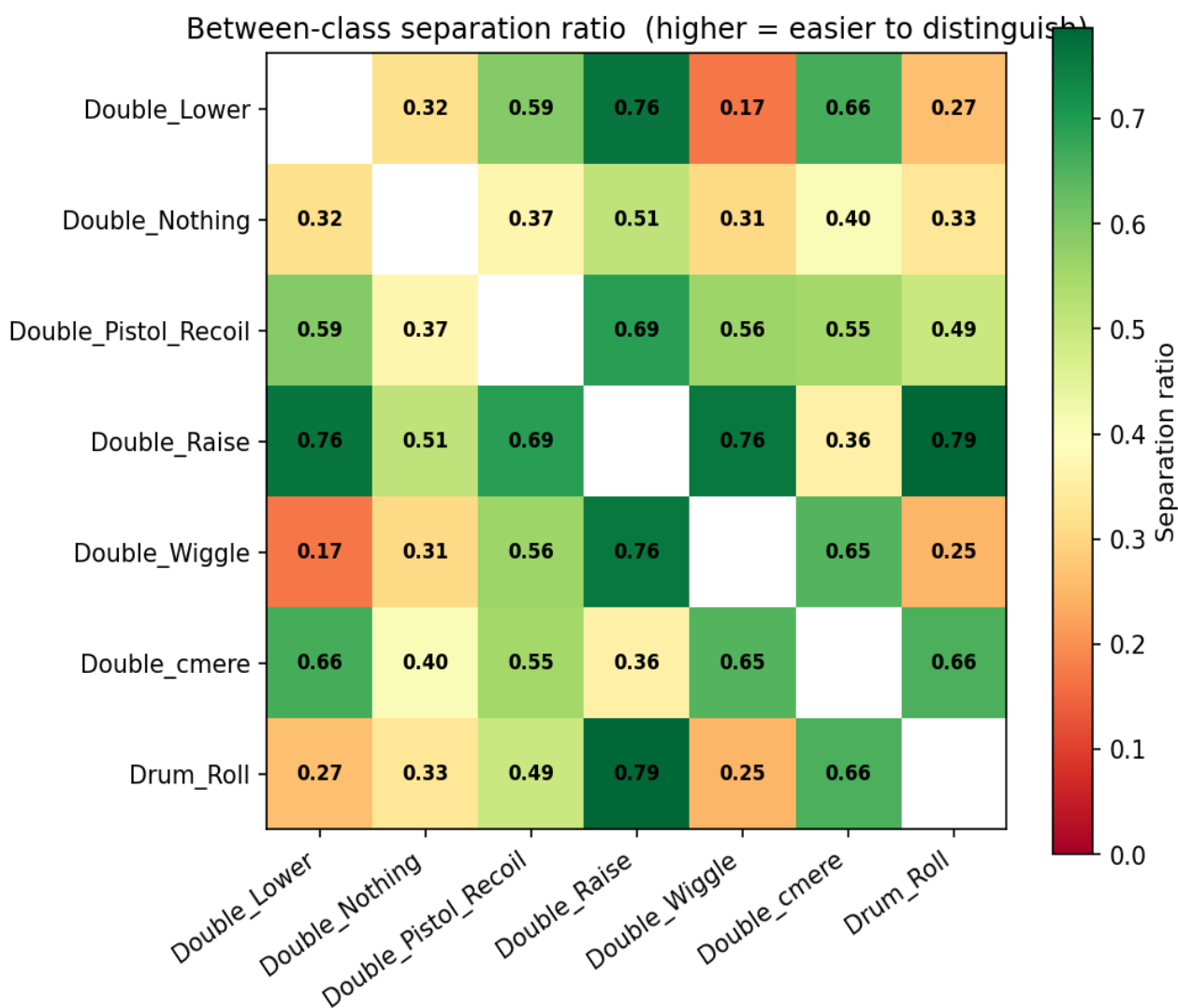
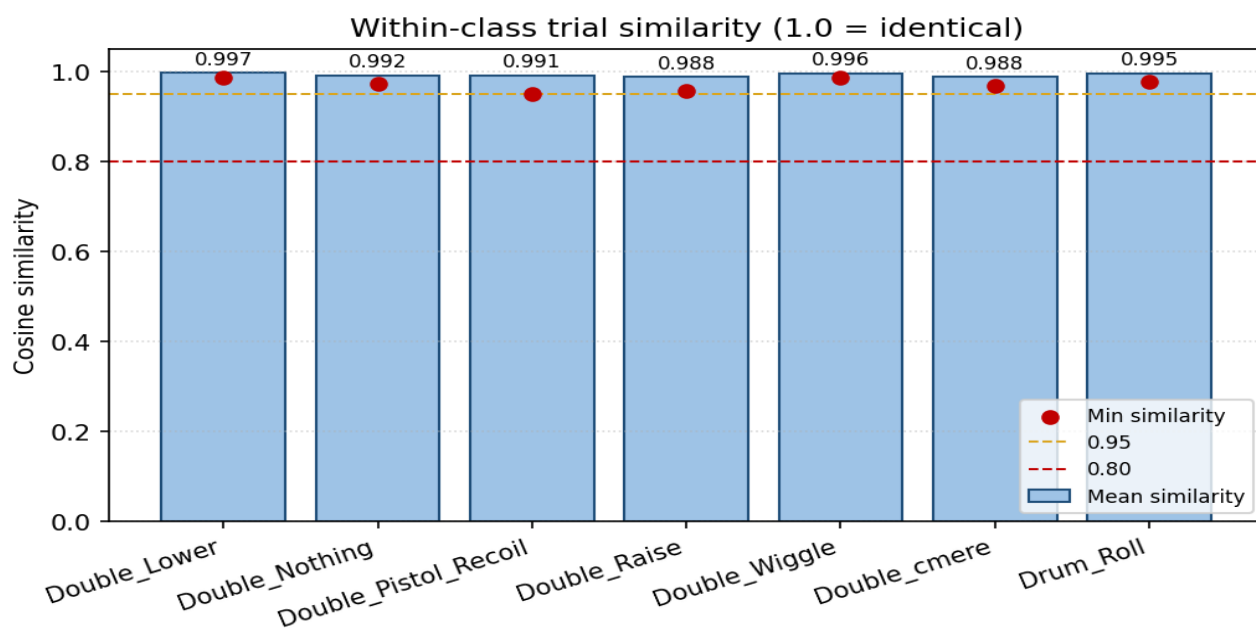


Mean training curves across LOUO folds

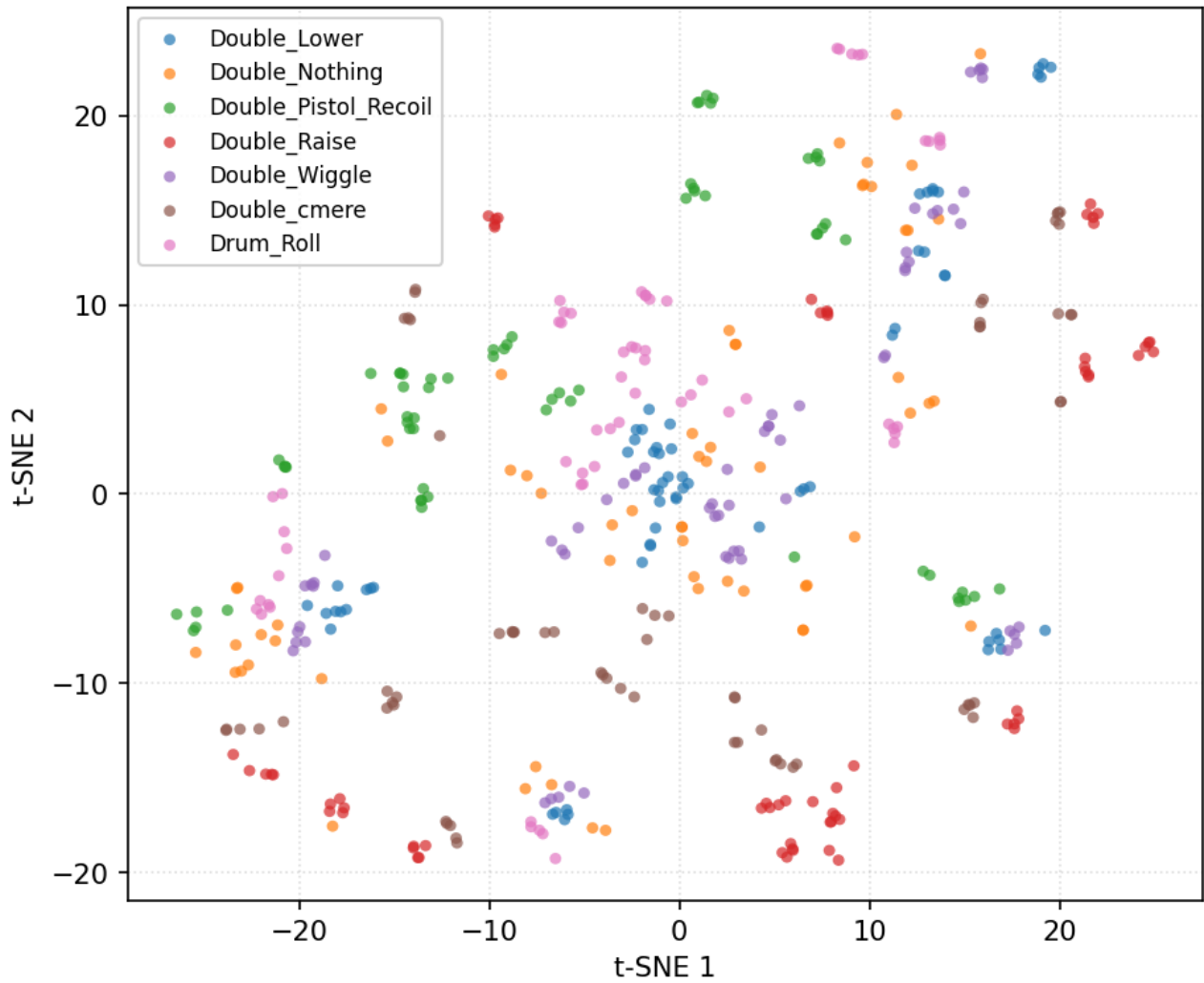


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



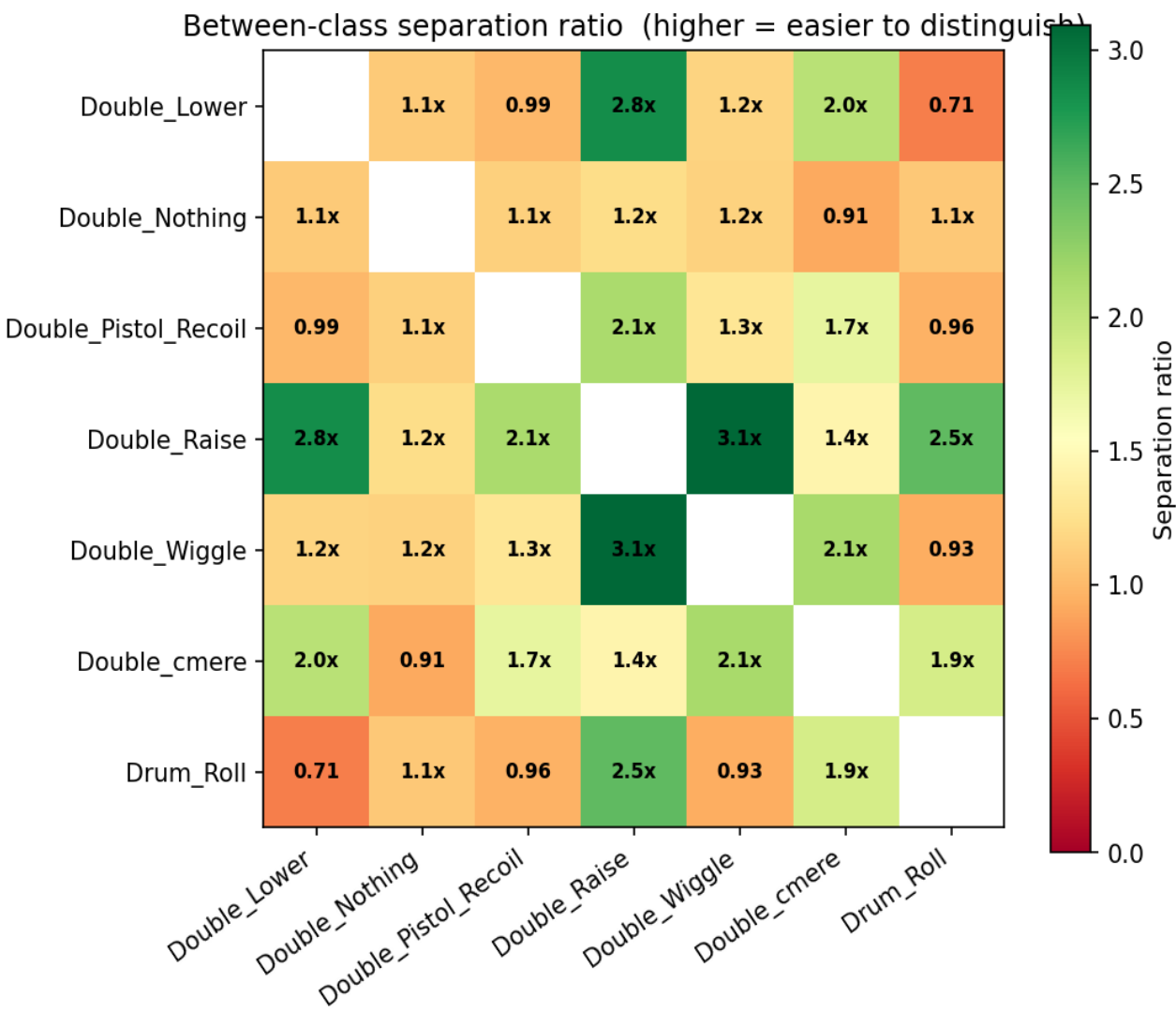
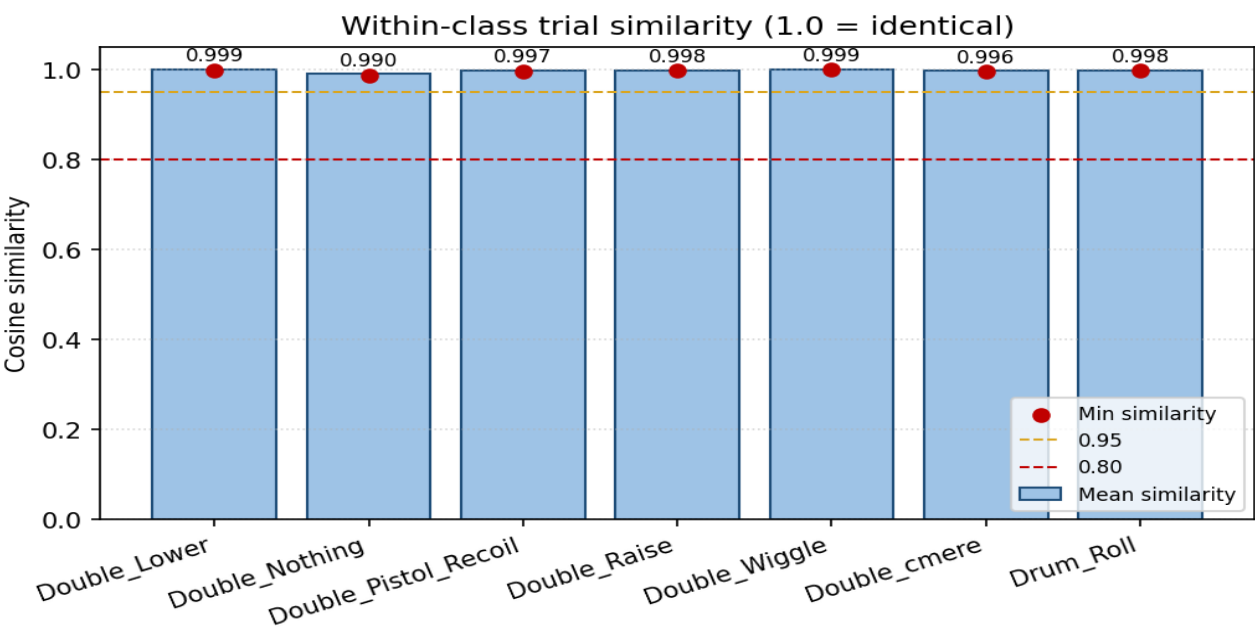
t-SNE feature space projection



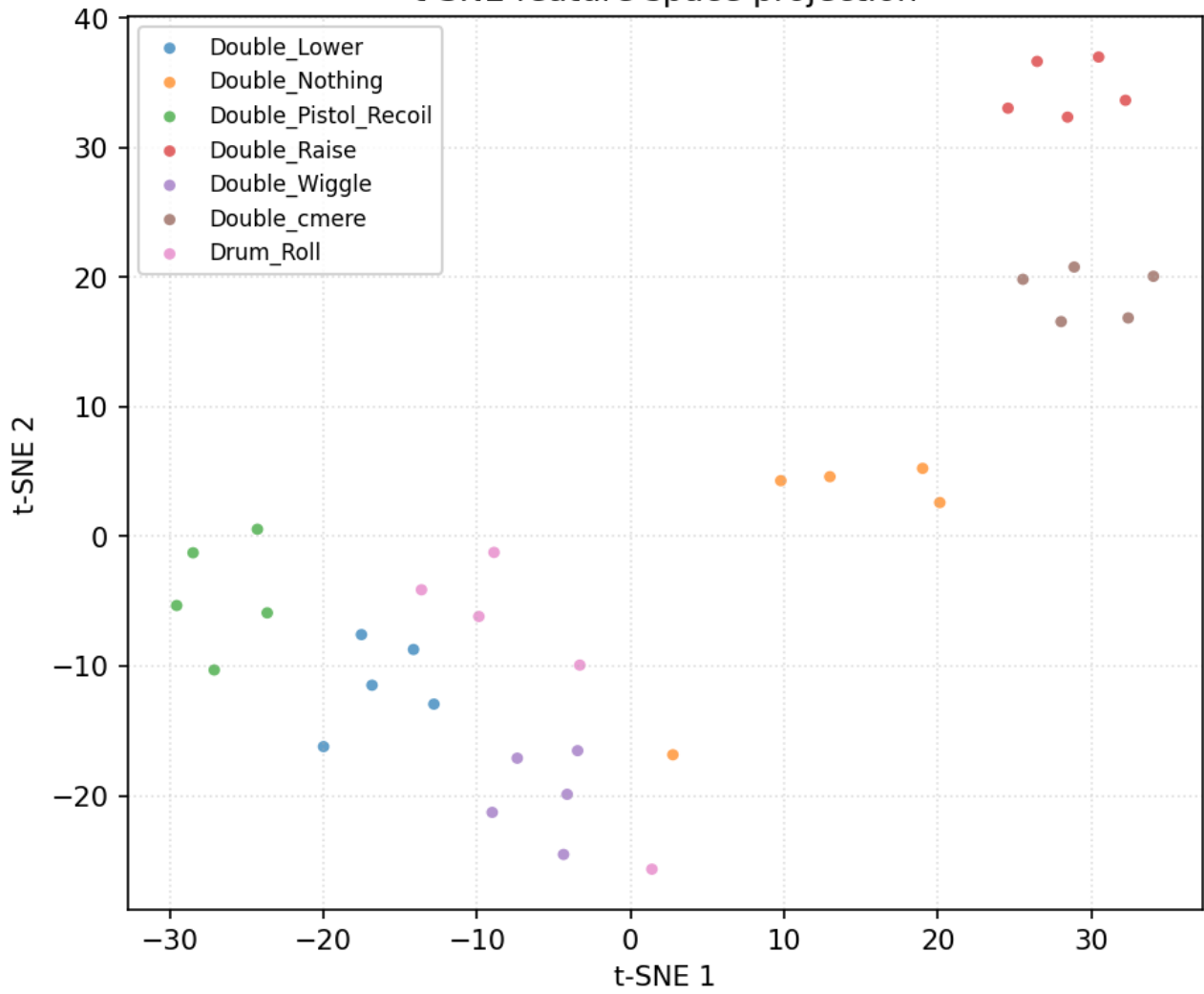
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

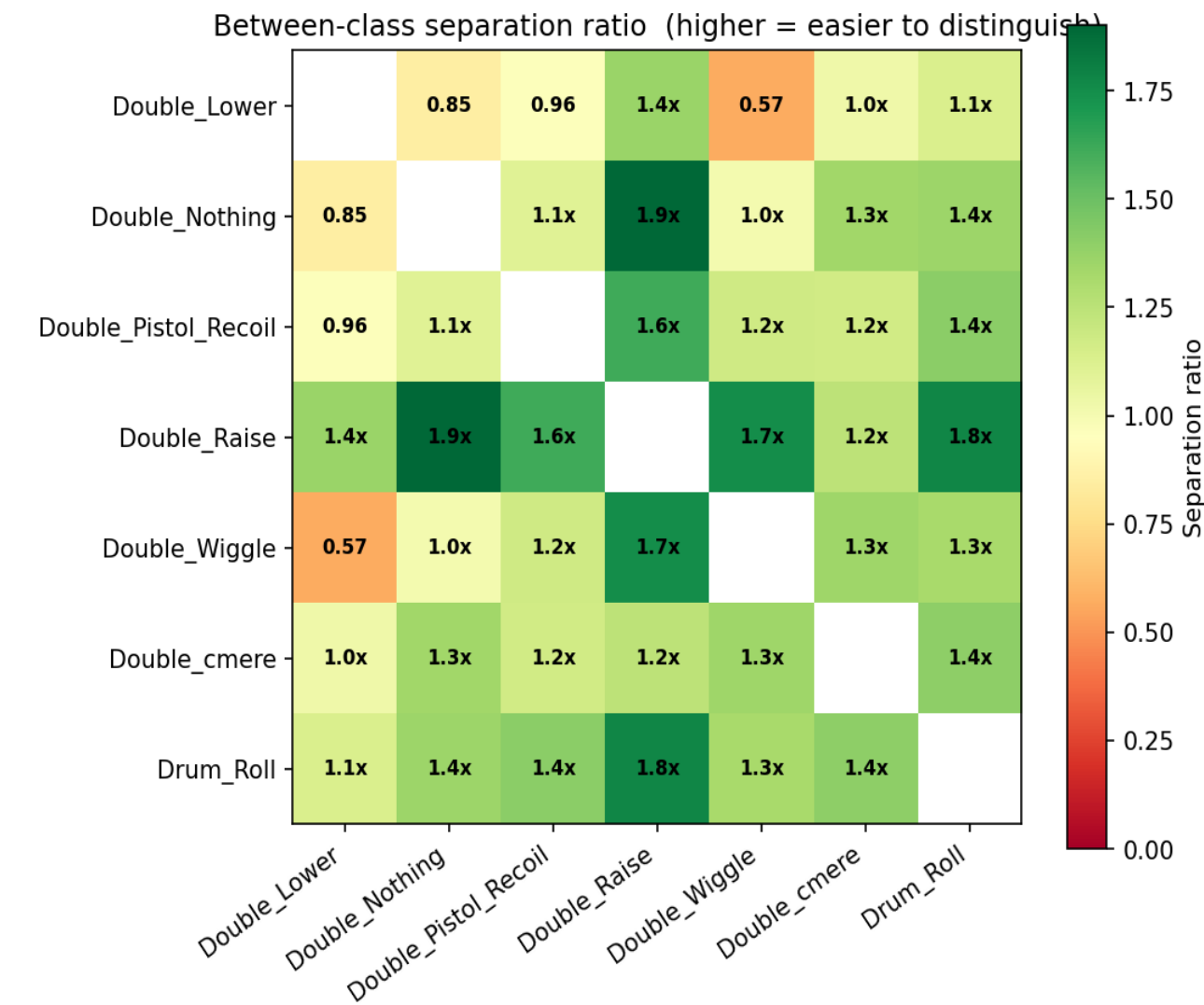
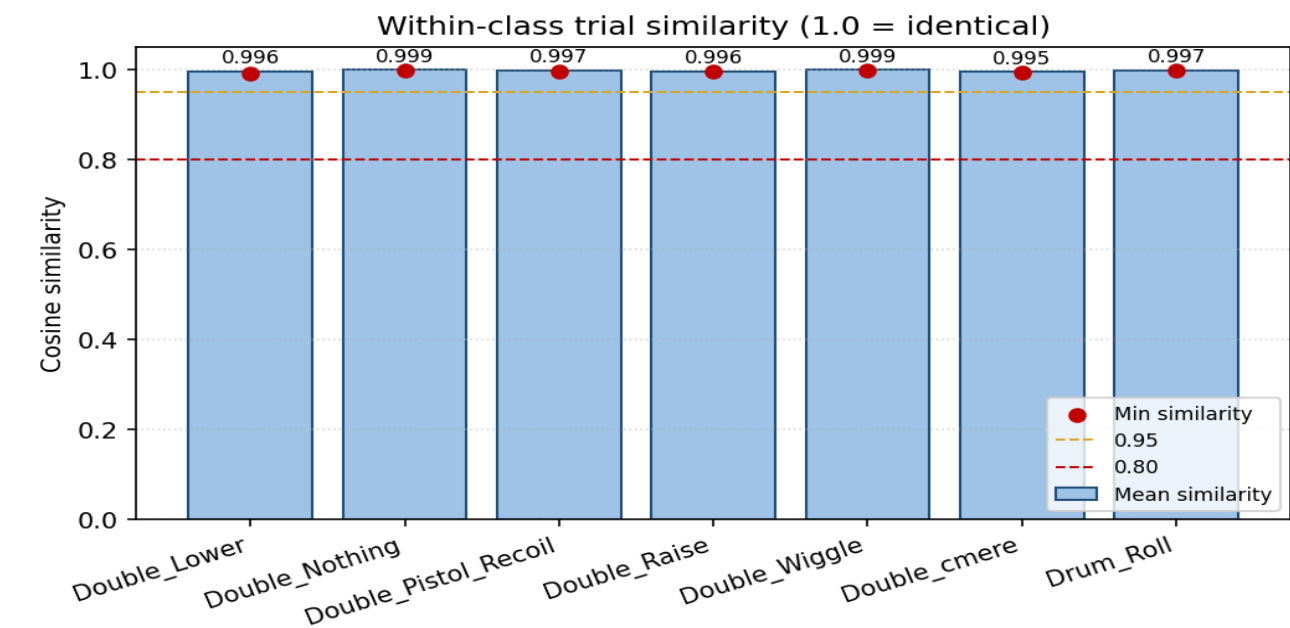


t-SNE feature space projection

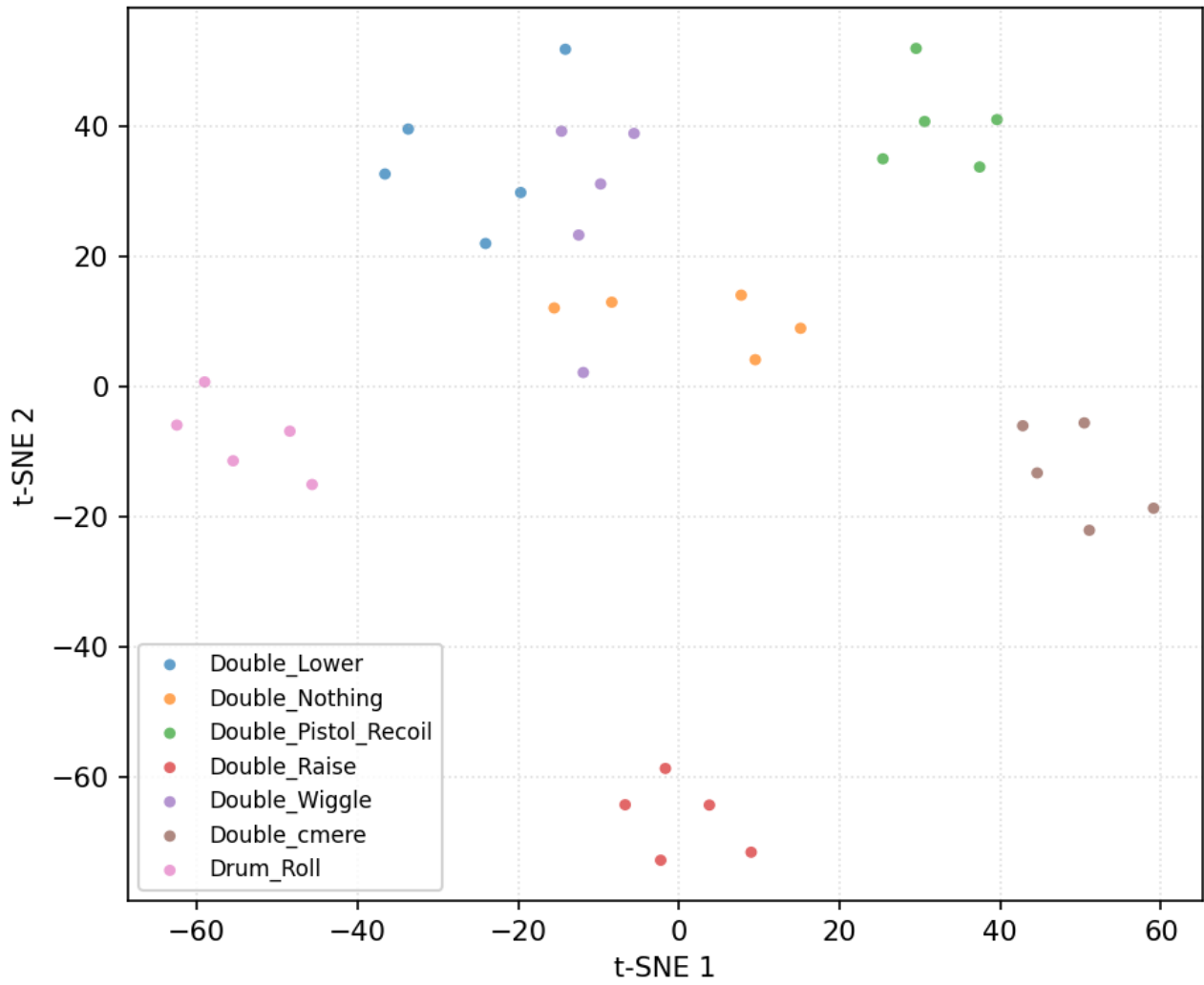


Data diagnostics — user 12_Marcus

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

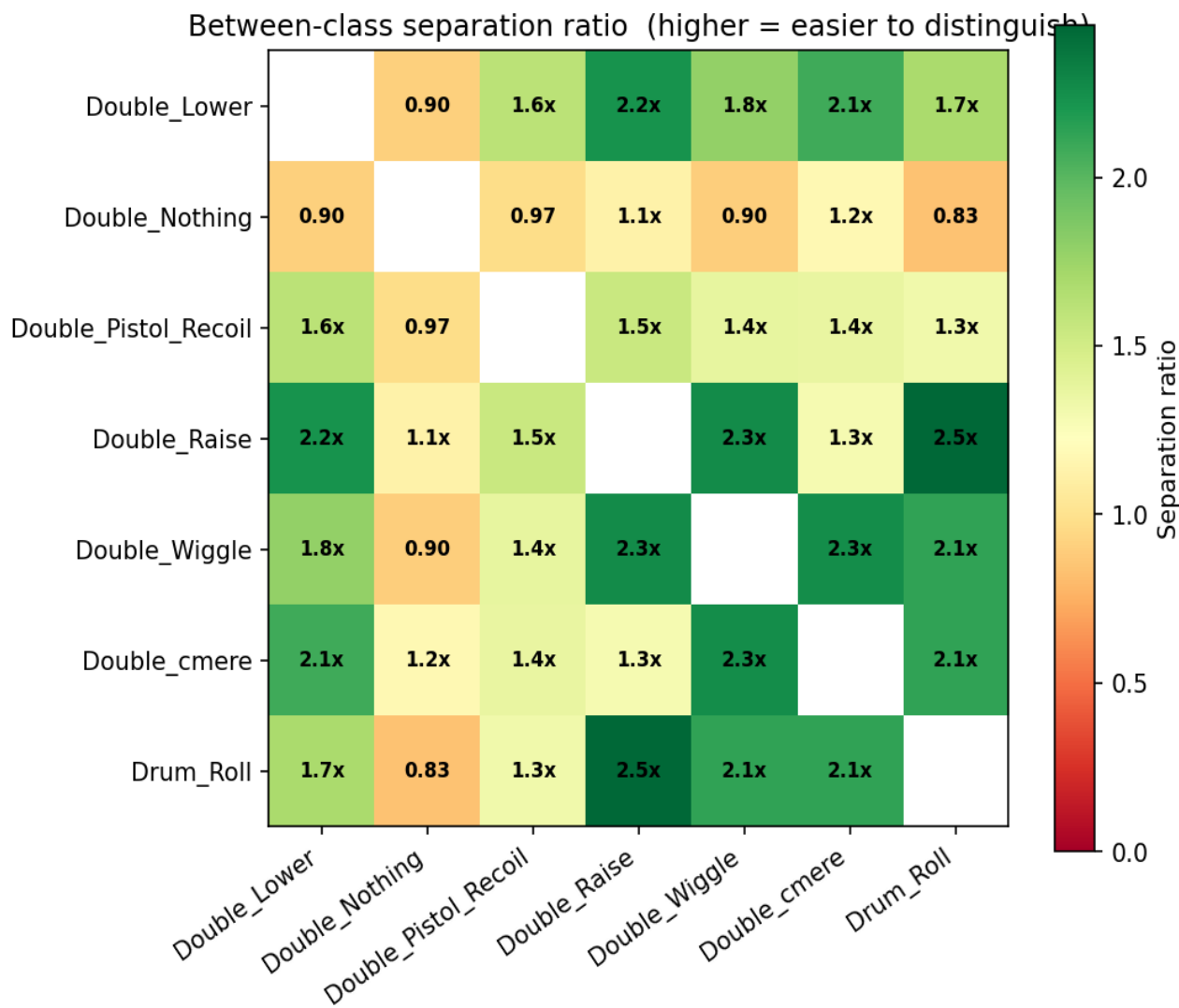
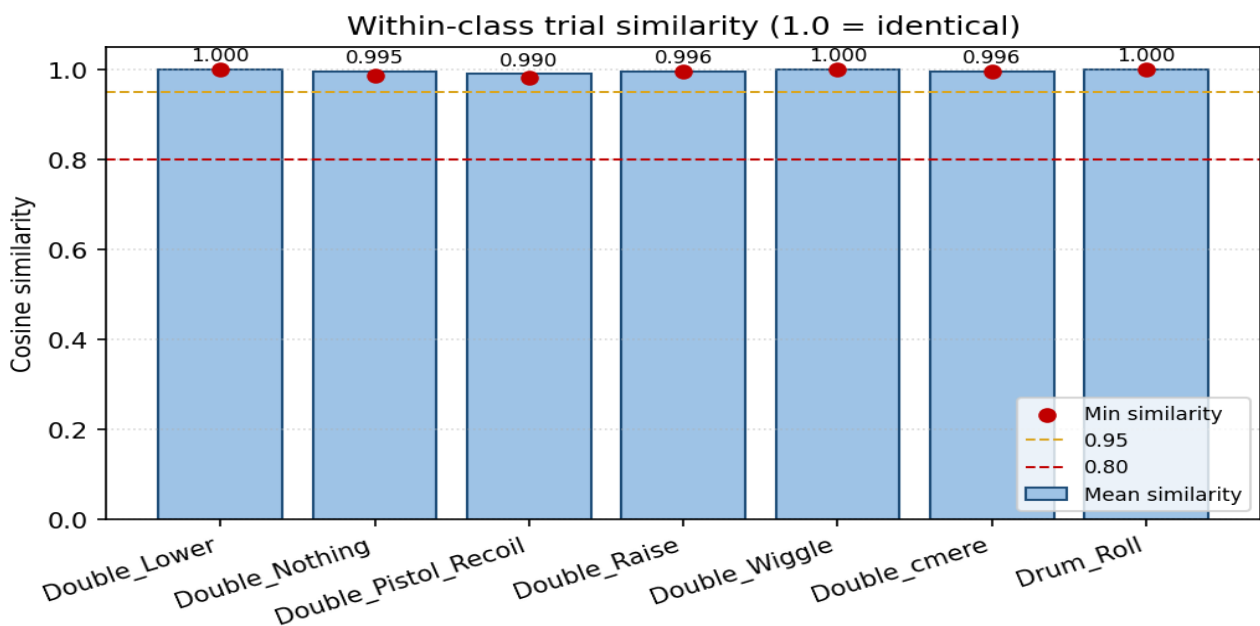


t-SNE feature space projection

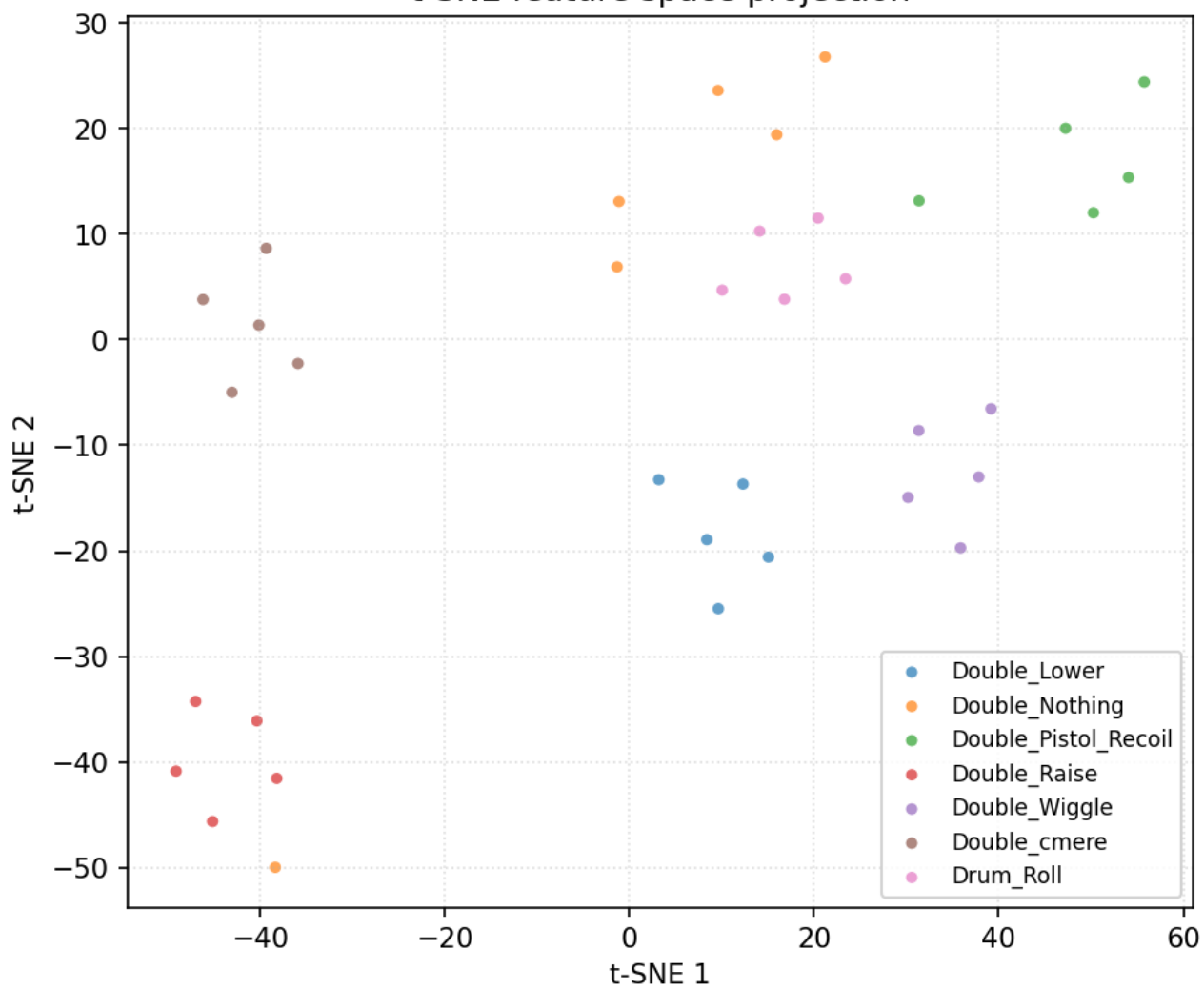


Data diagnostics — user 14_StephenV2

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

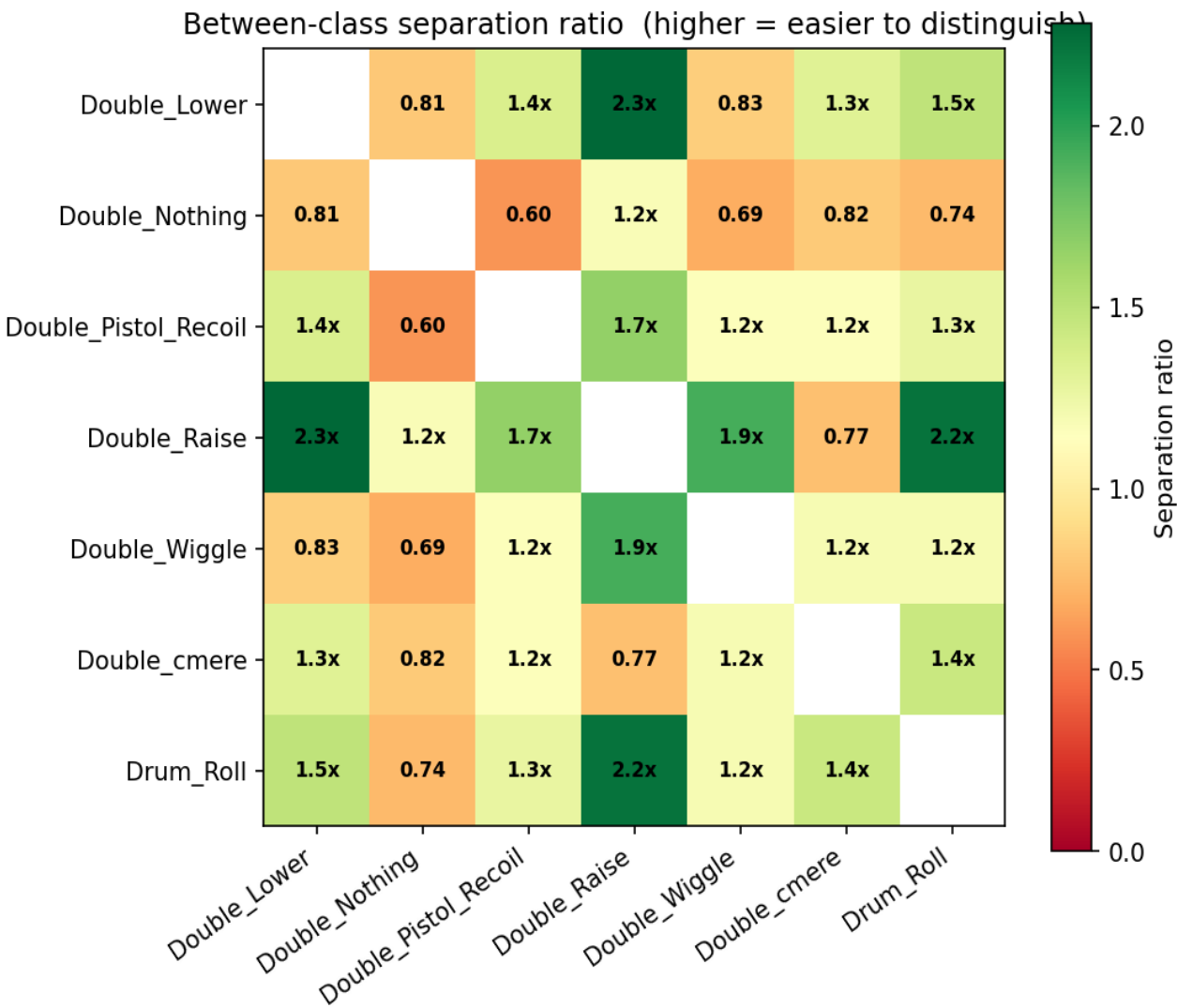
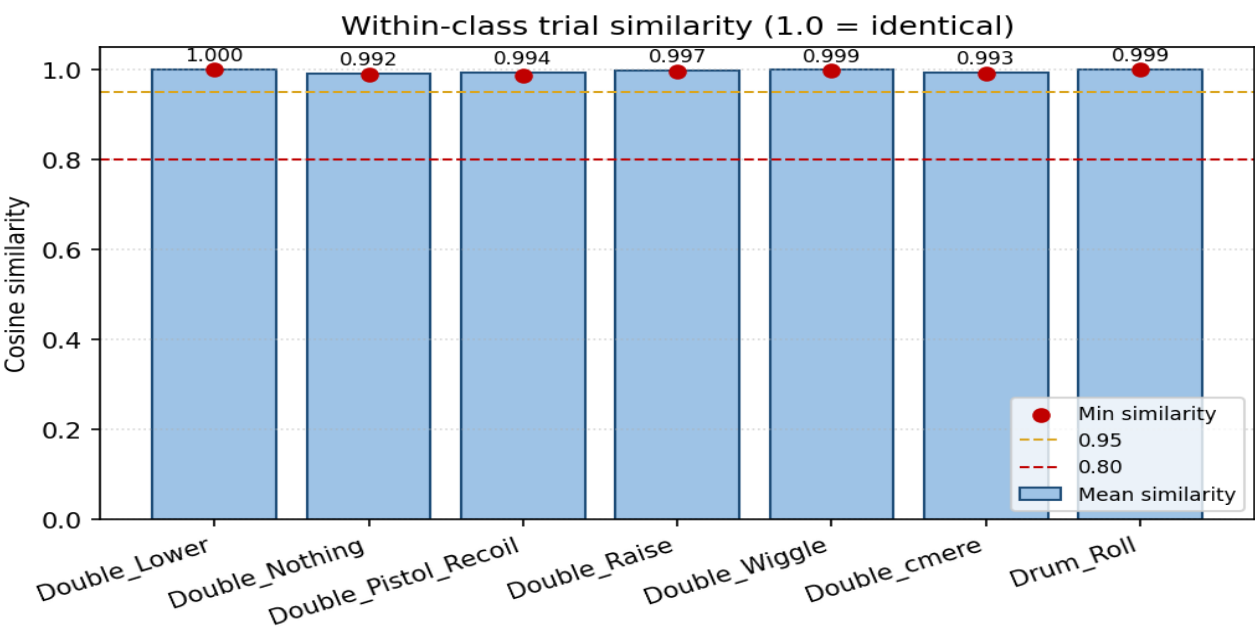


t-SNE feature space projection

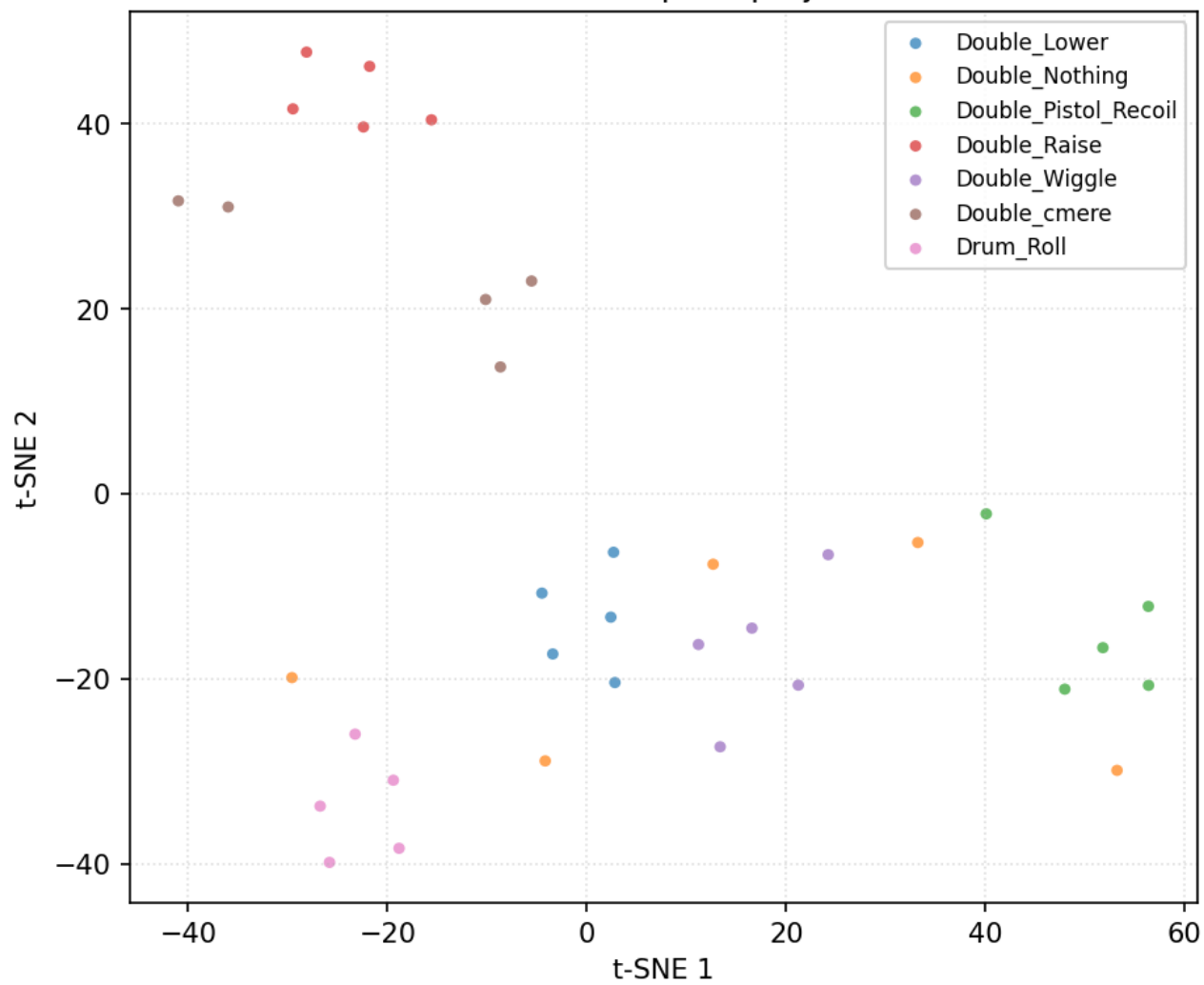


Data diagnostics — user 15_Tash

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

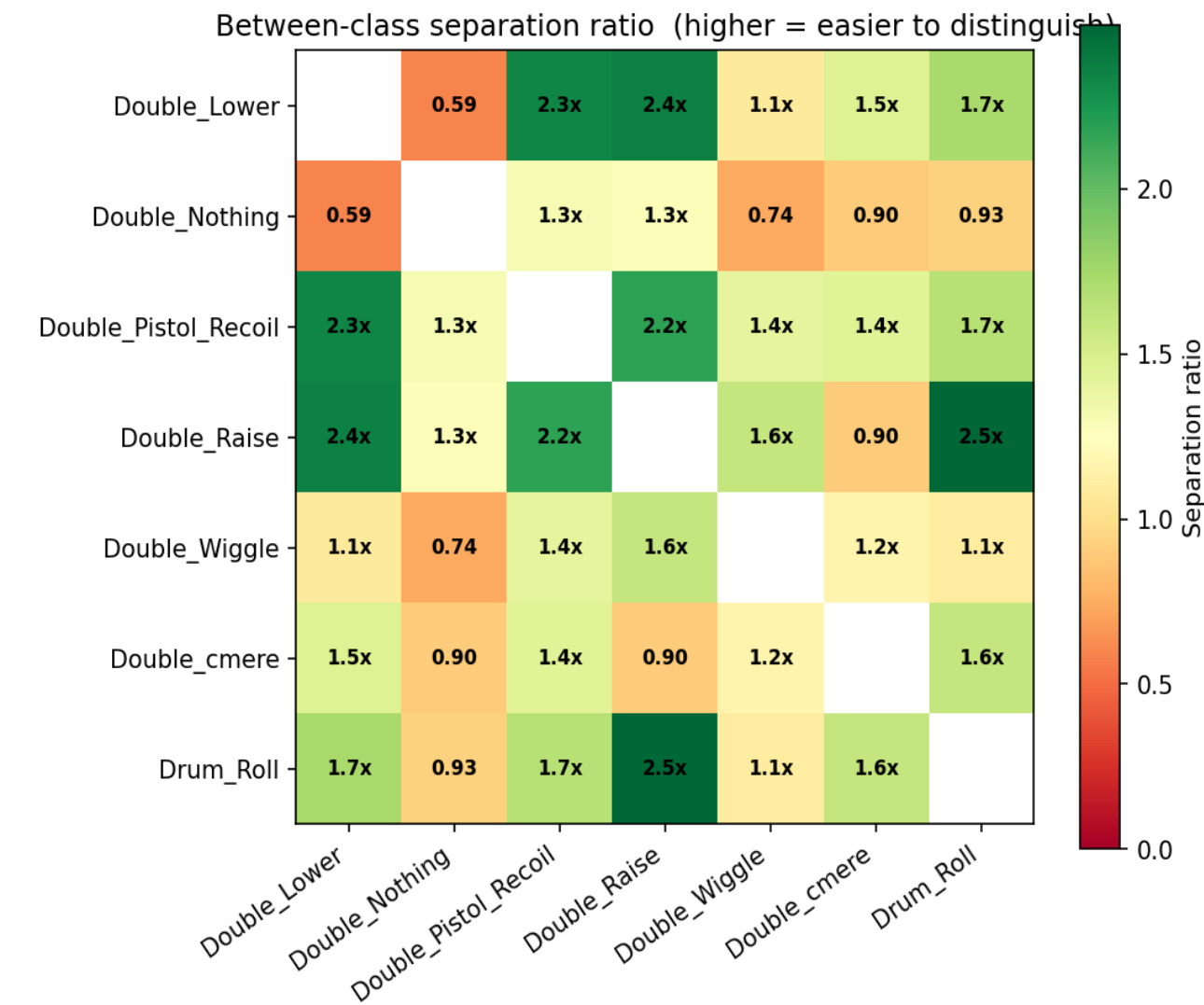
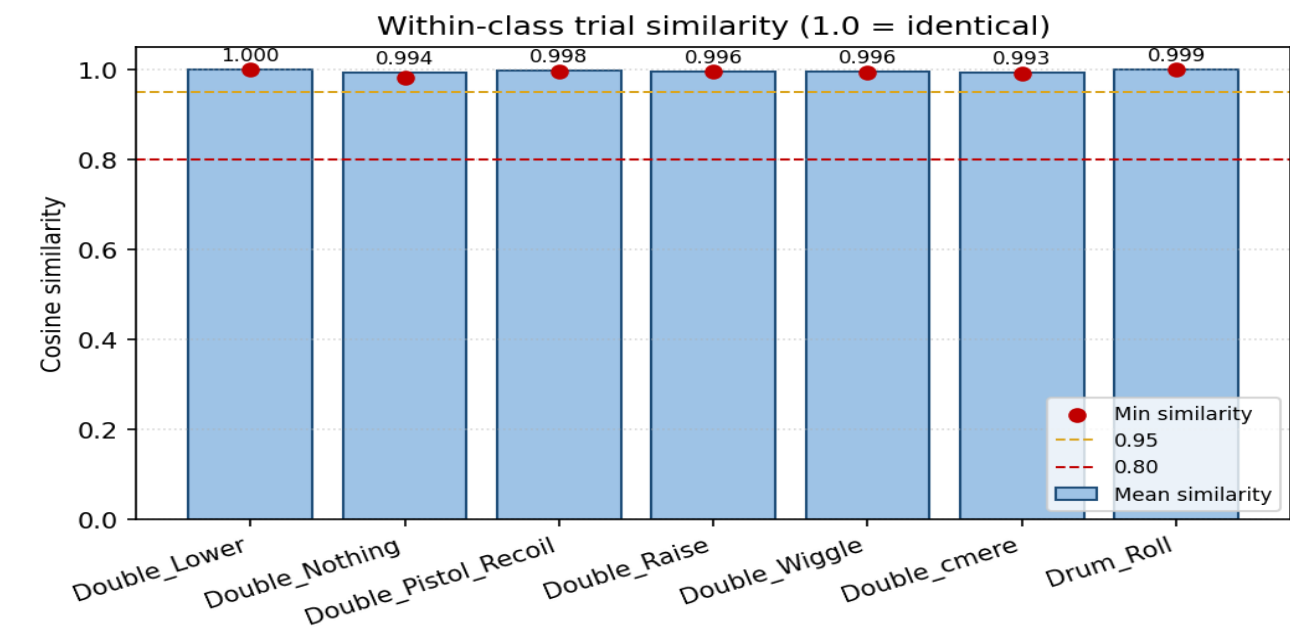


t-SNE feature space projection

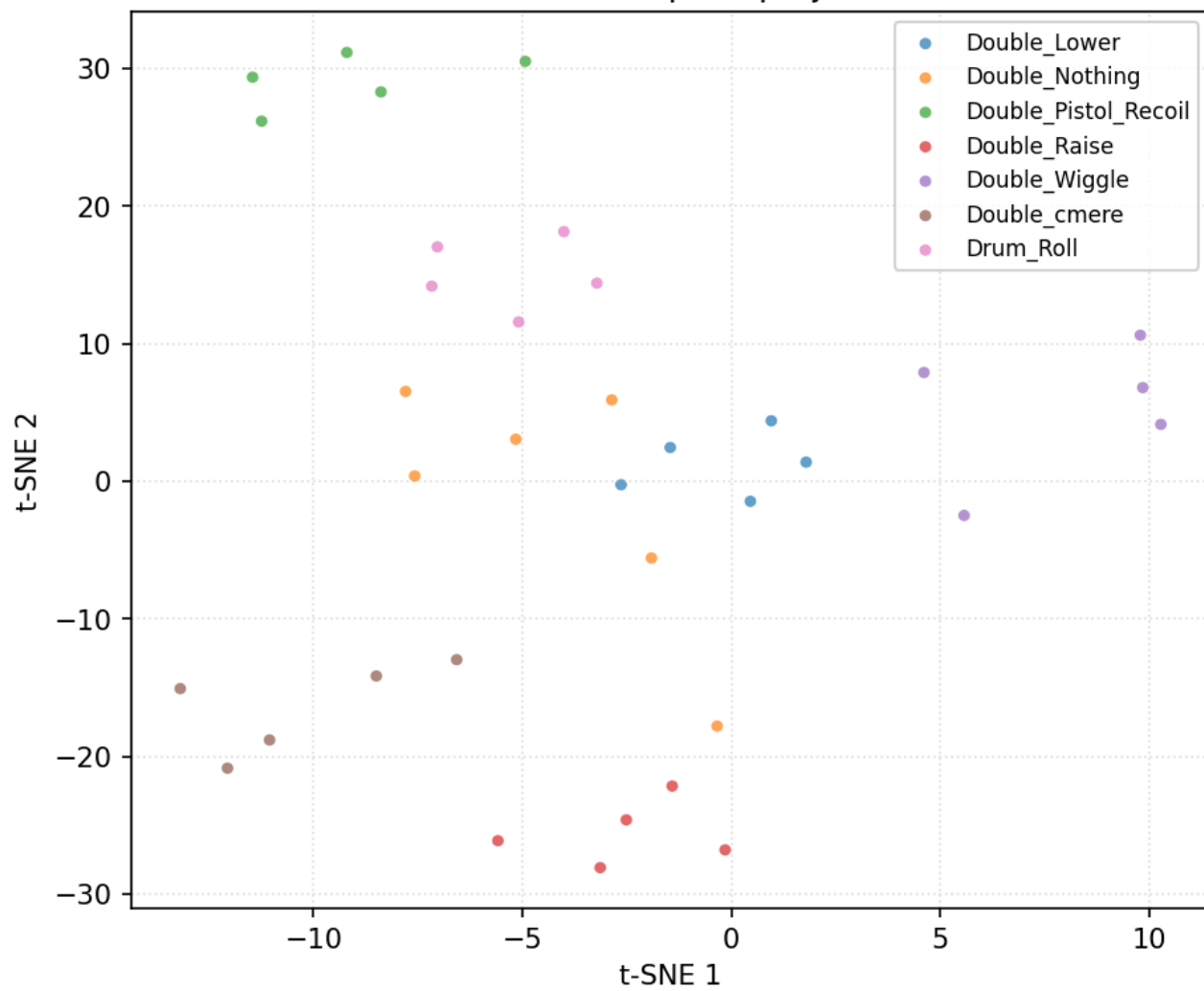


Data diagnostics — user 16_Bella

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

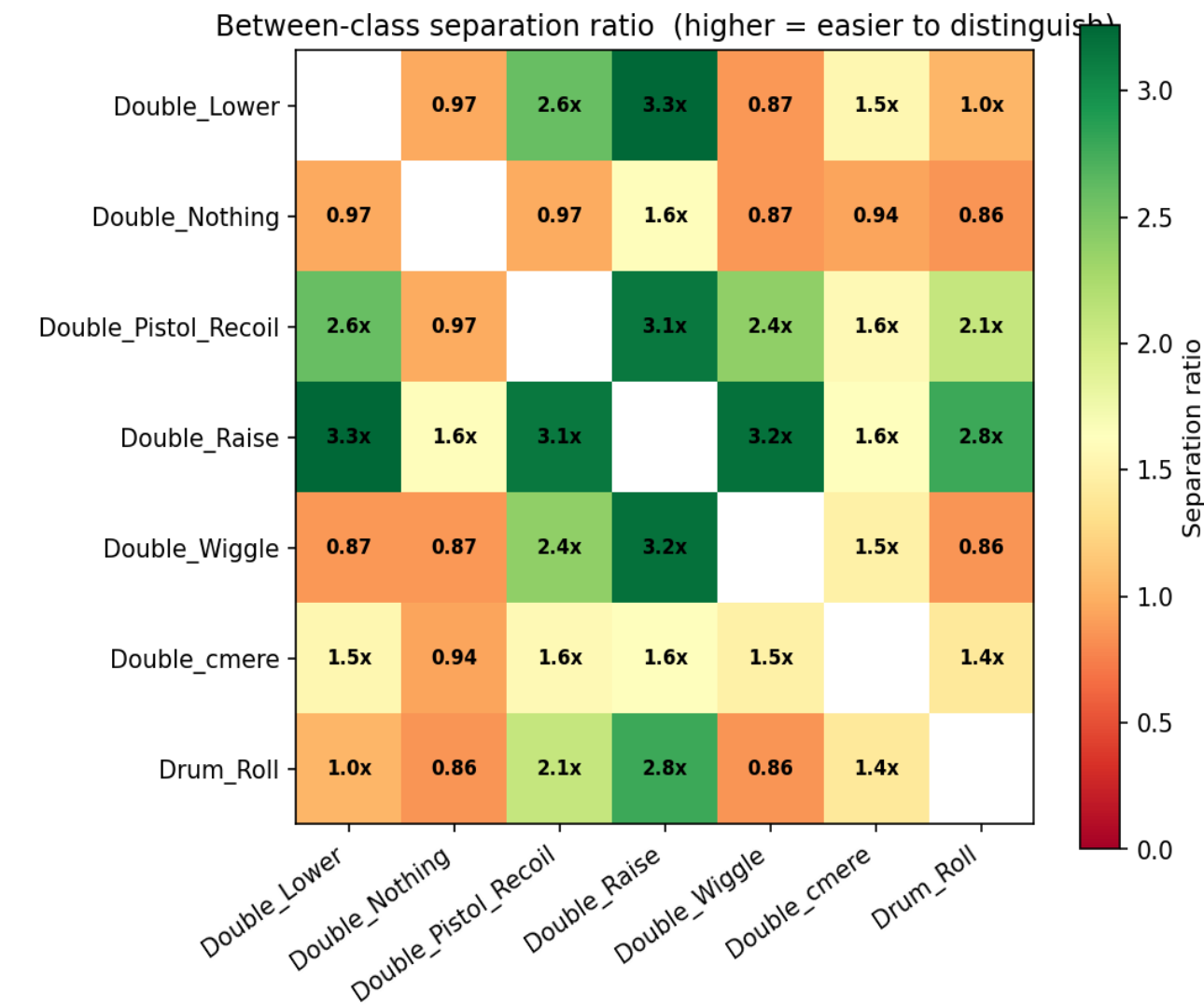
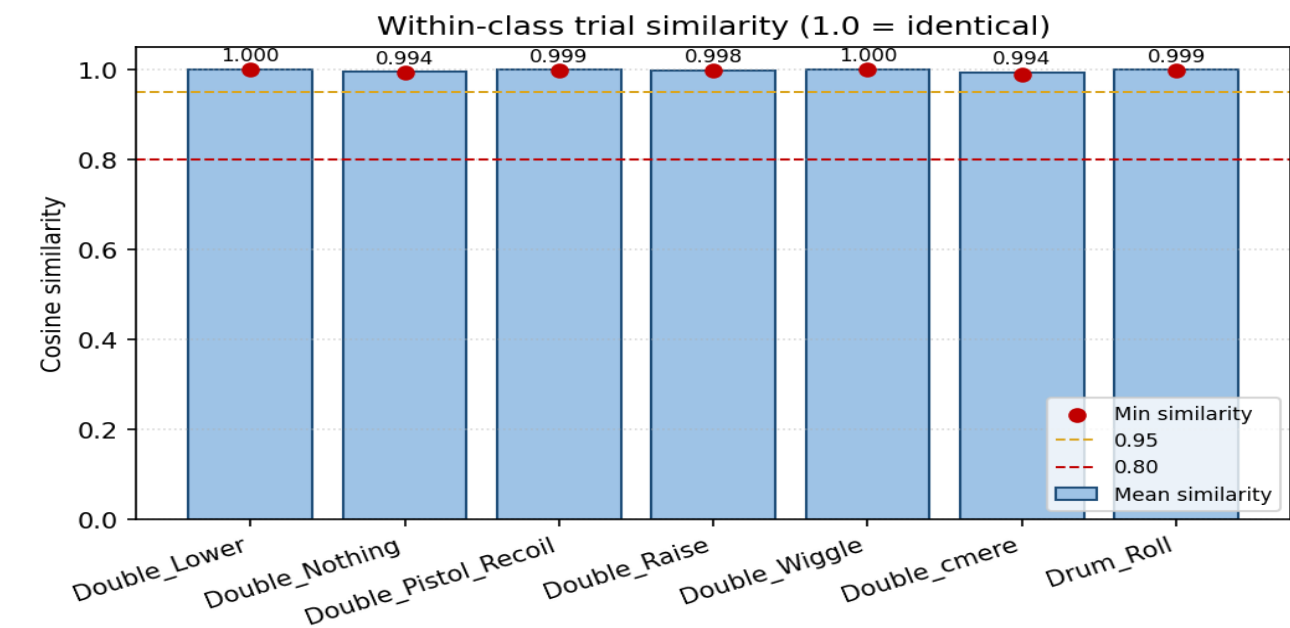


t-SNE feature space projection

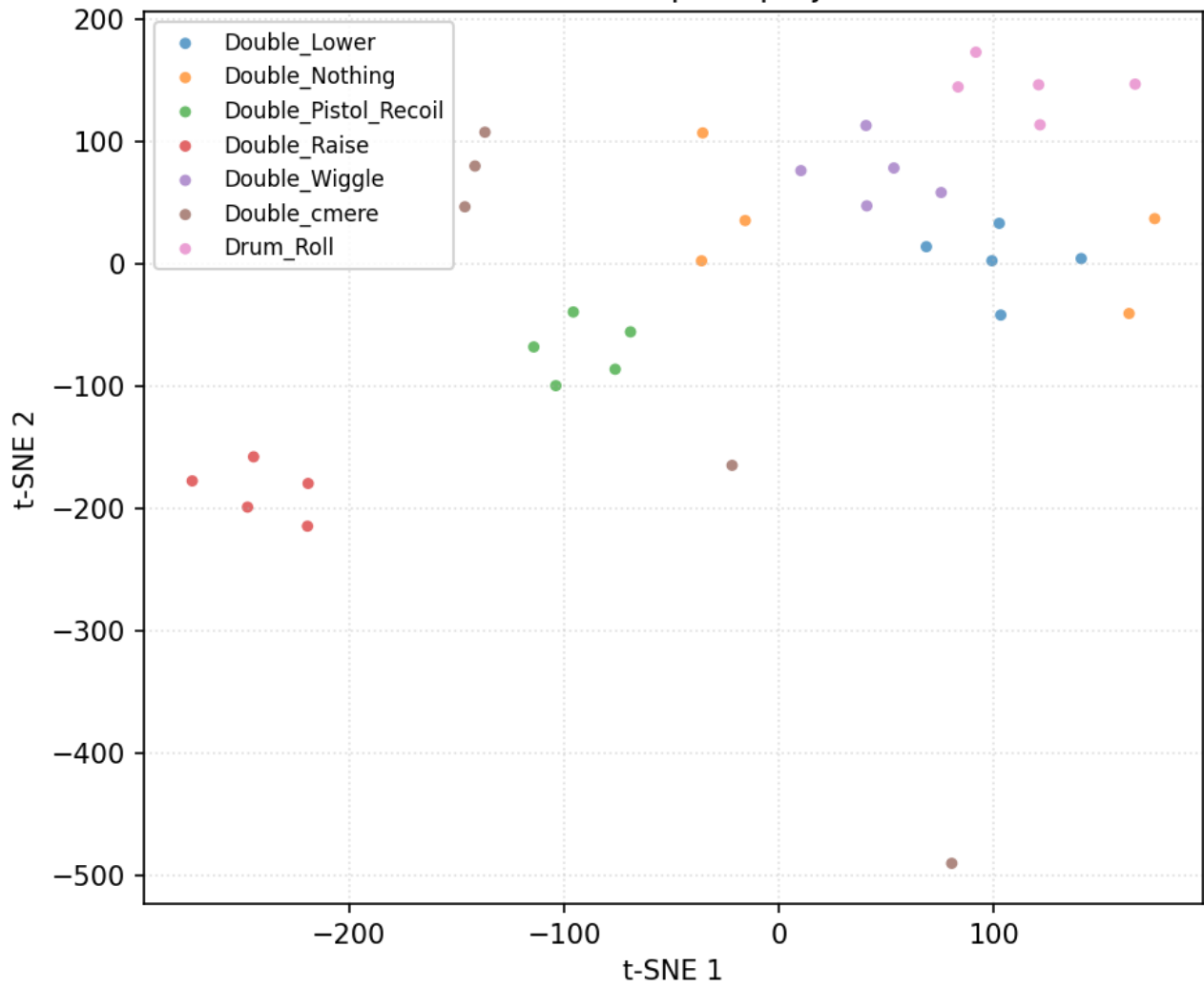


Data diagnostics — user 17_Raquel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

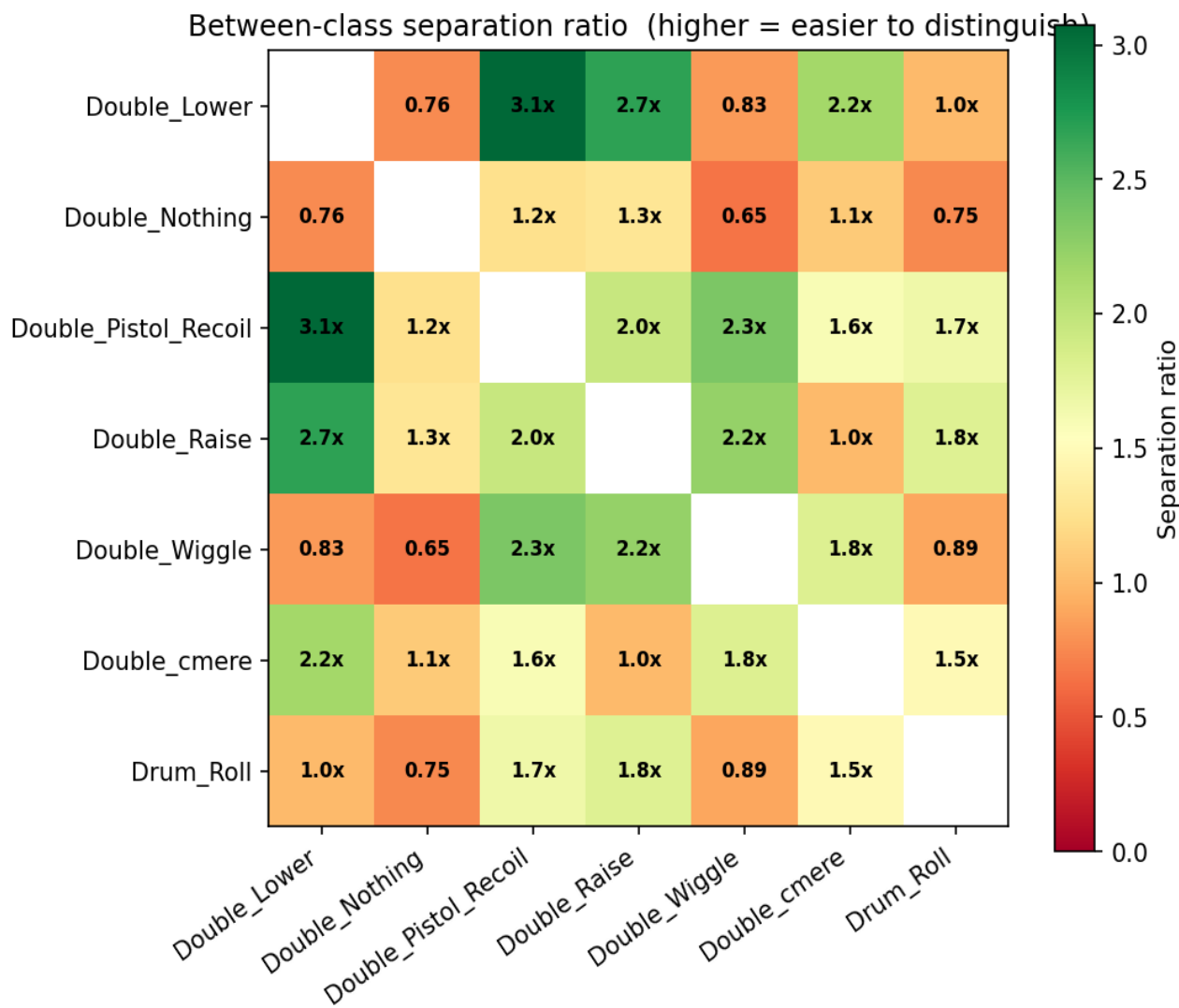
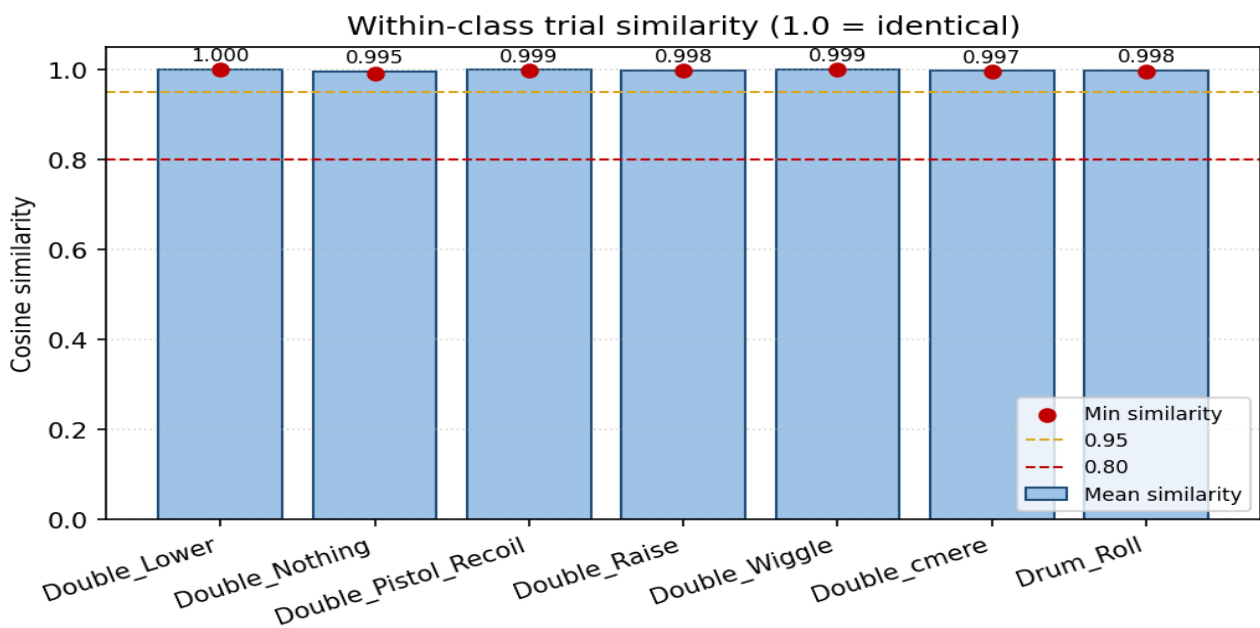


t-SNE feature space projection

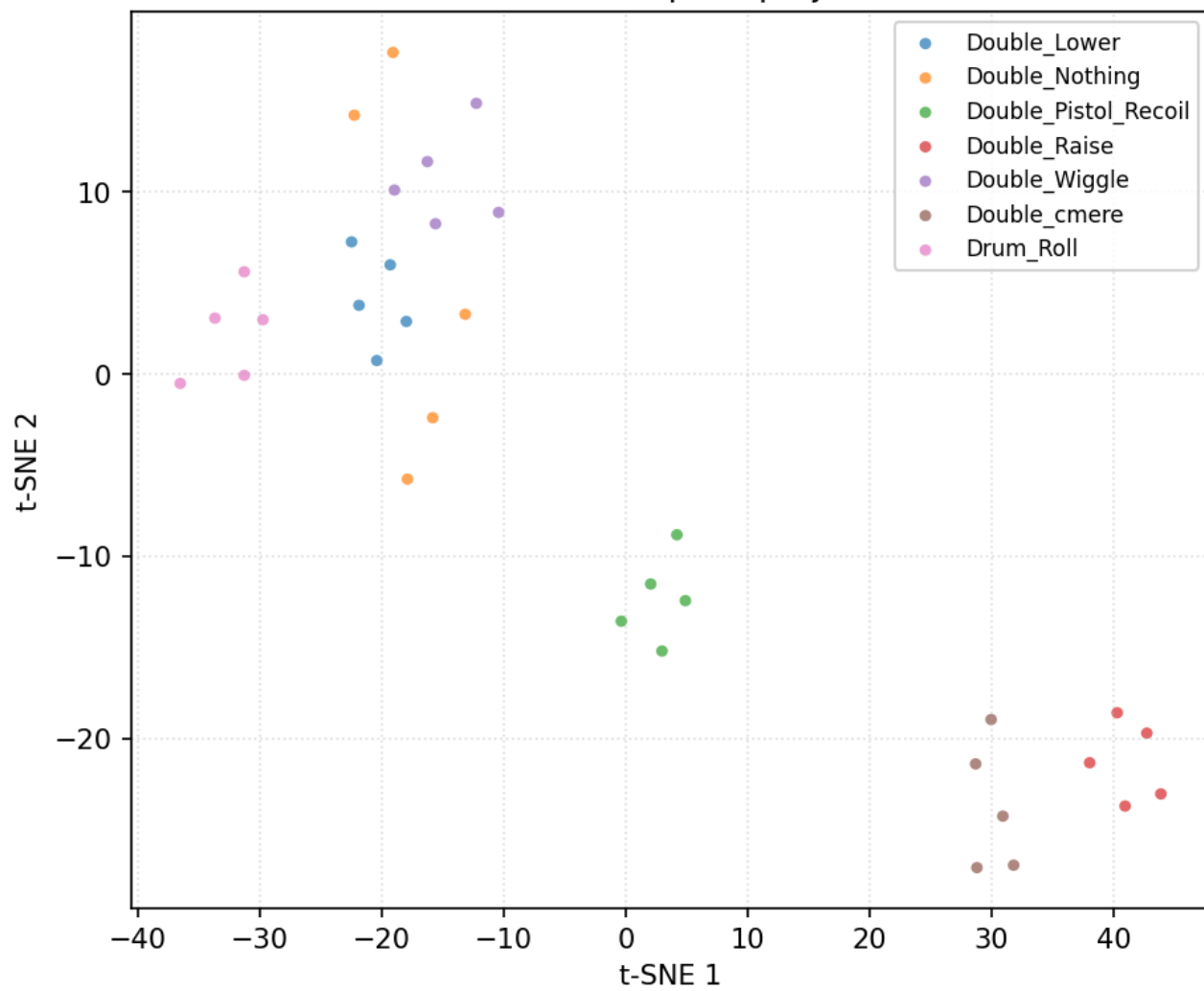


Data diagnostics — user 18_Bridgette

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

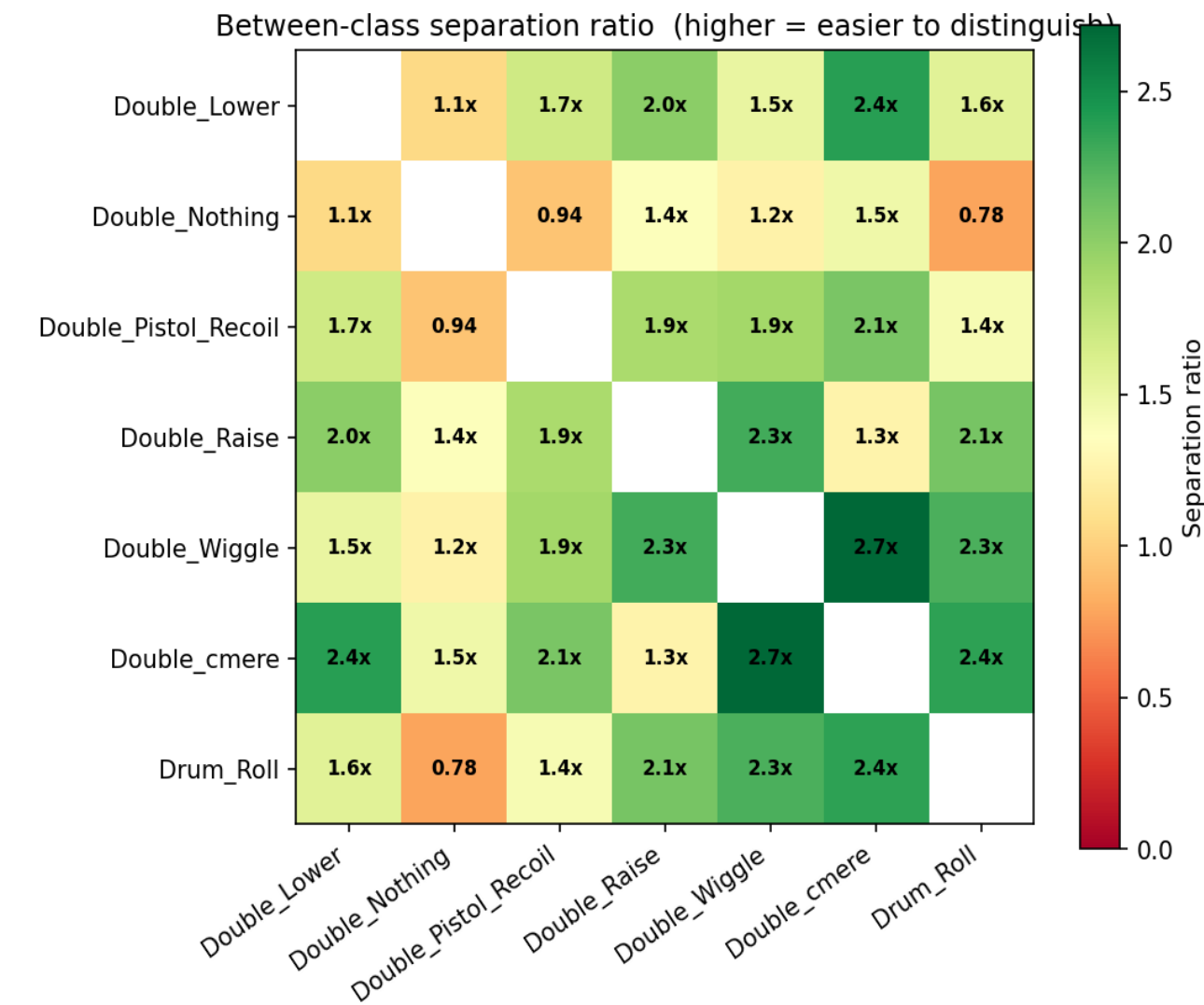
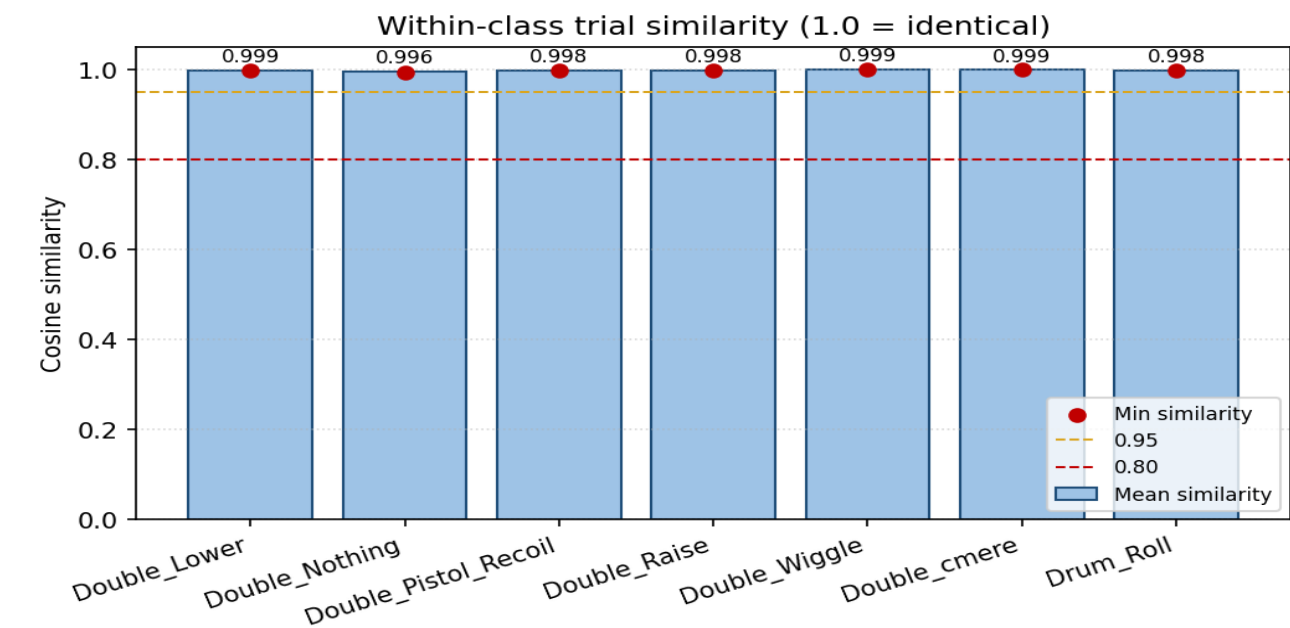


t-SNE feature space projection

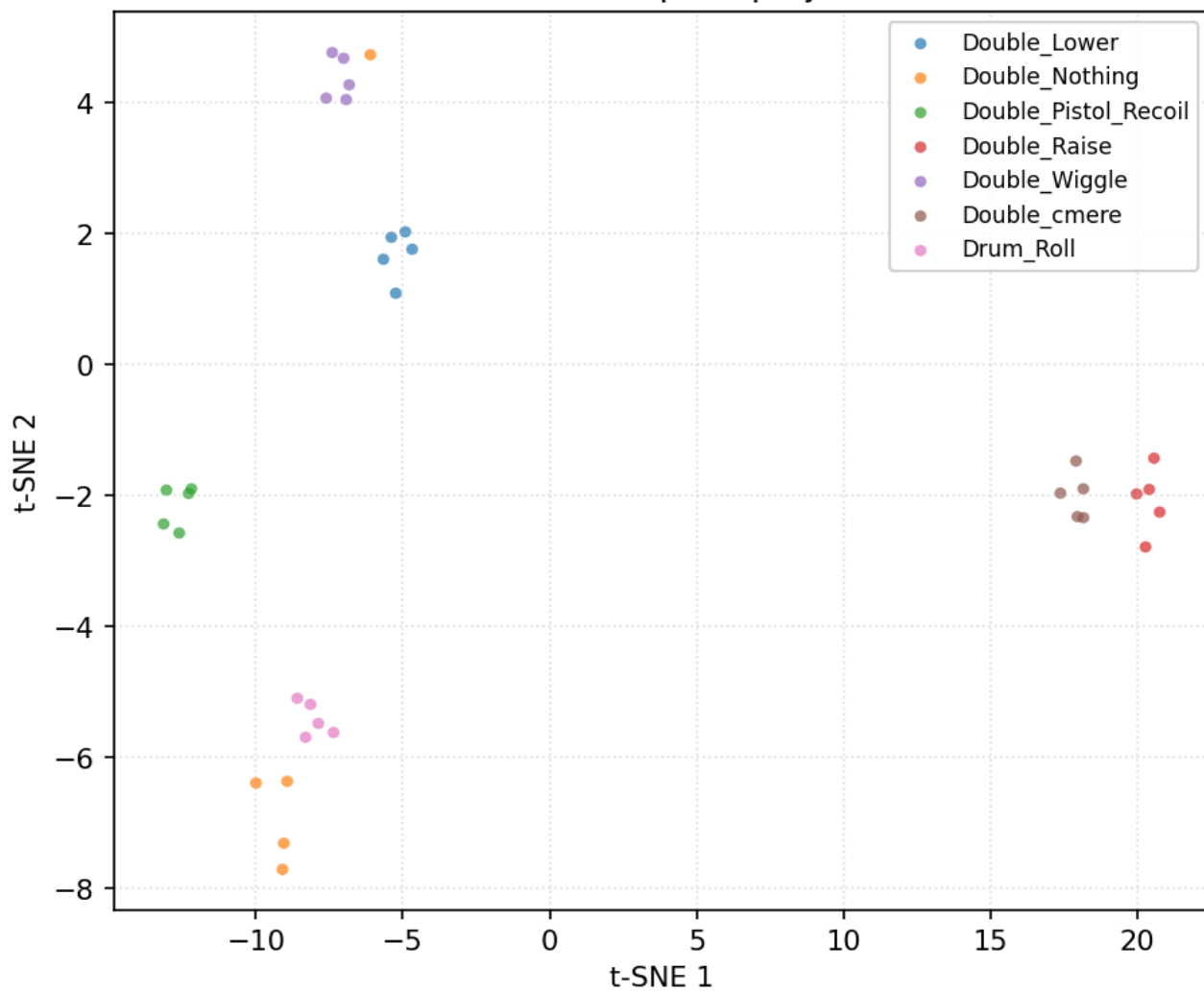


Data diagnostics — user 1_Alán

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

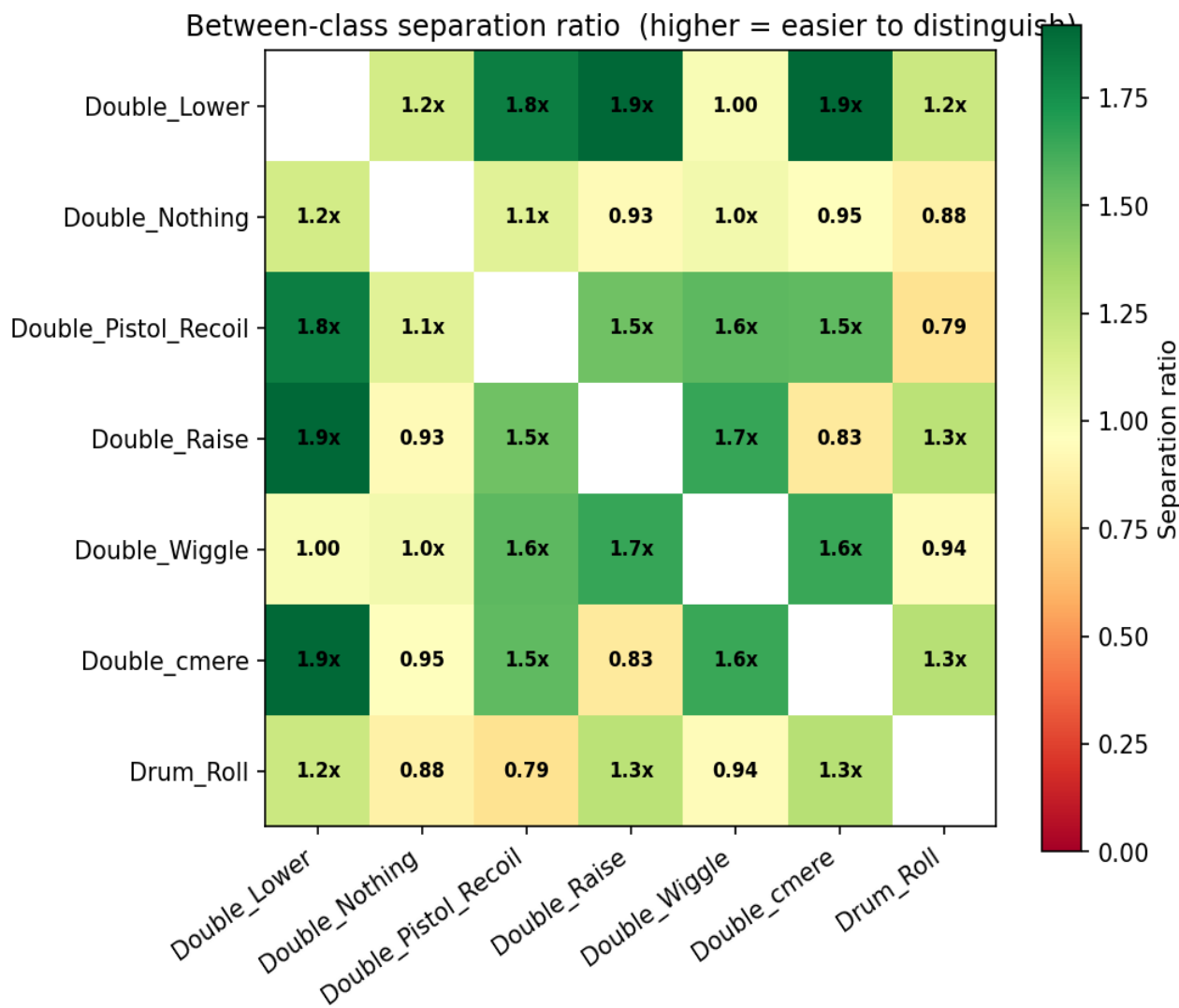
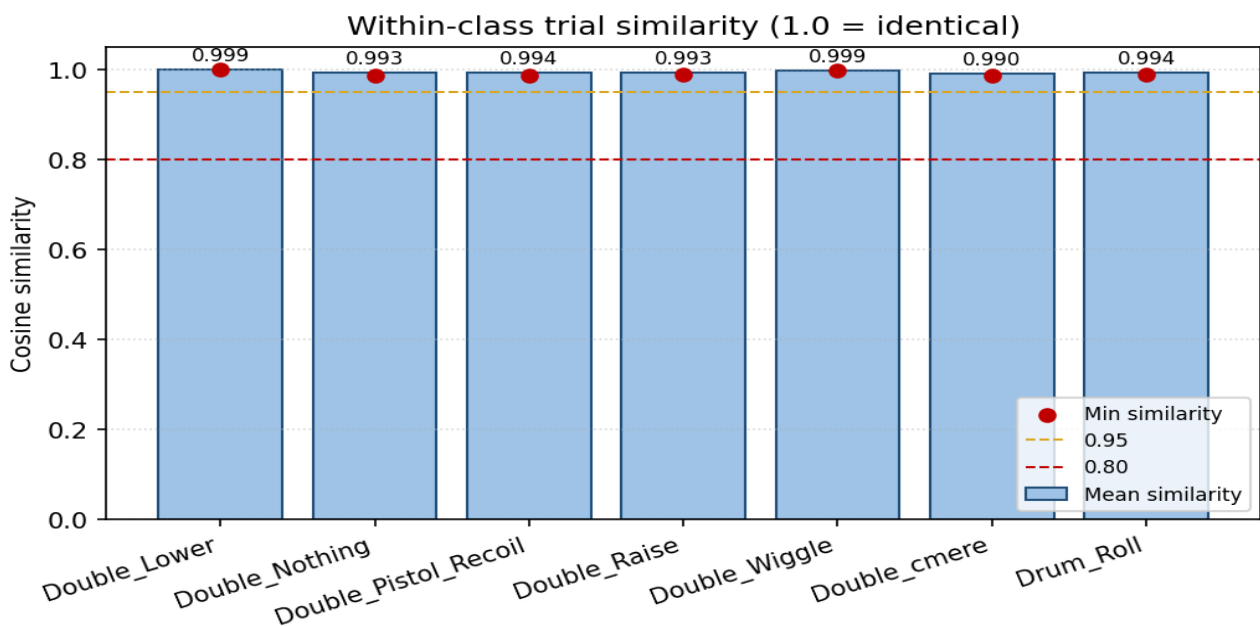


t-SNE feature space projection

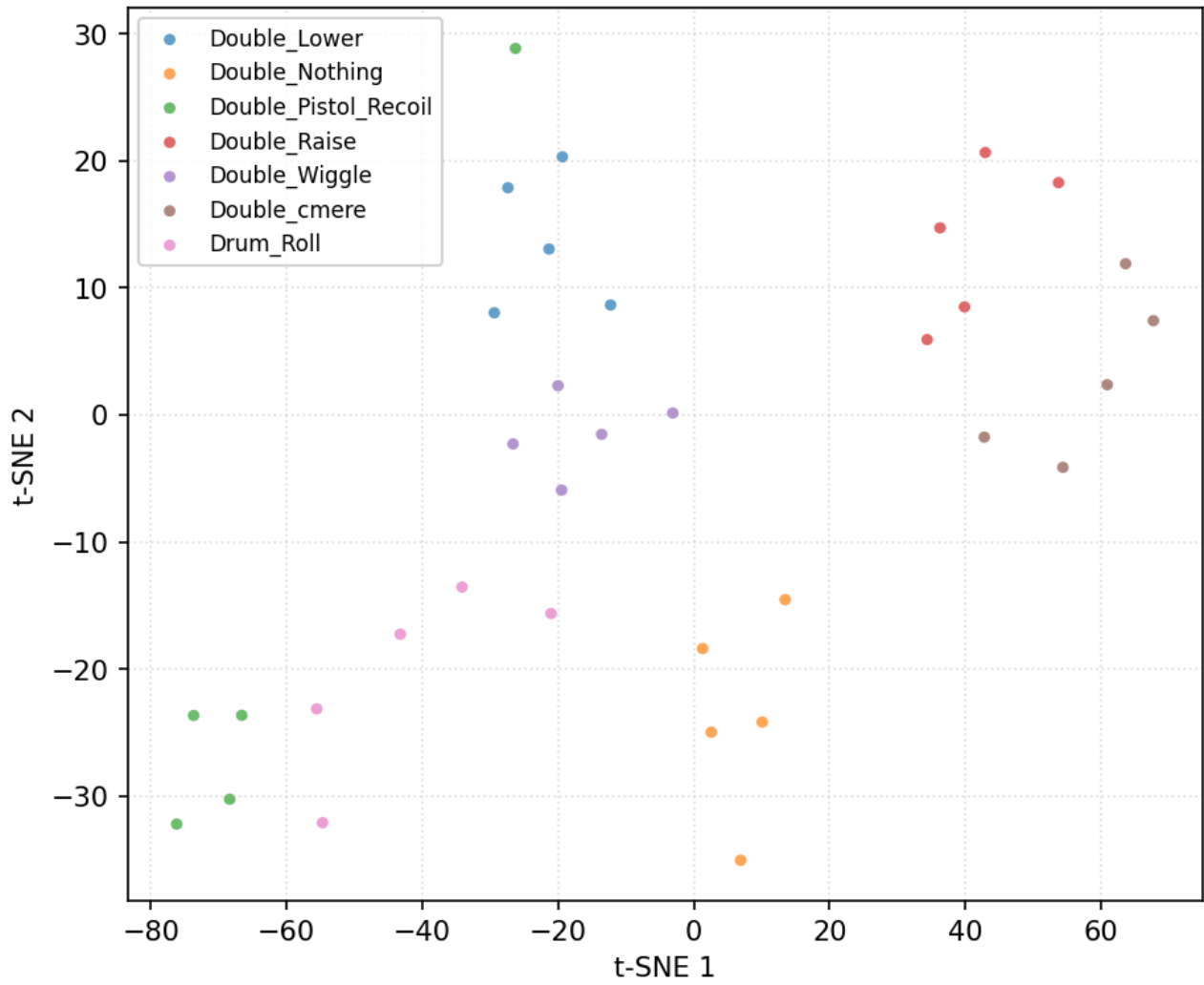


Data diagnostics — user 2_Alex

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

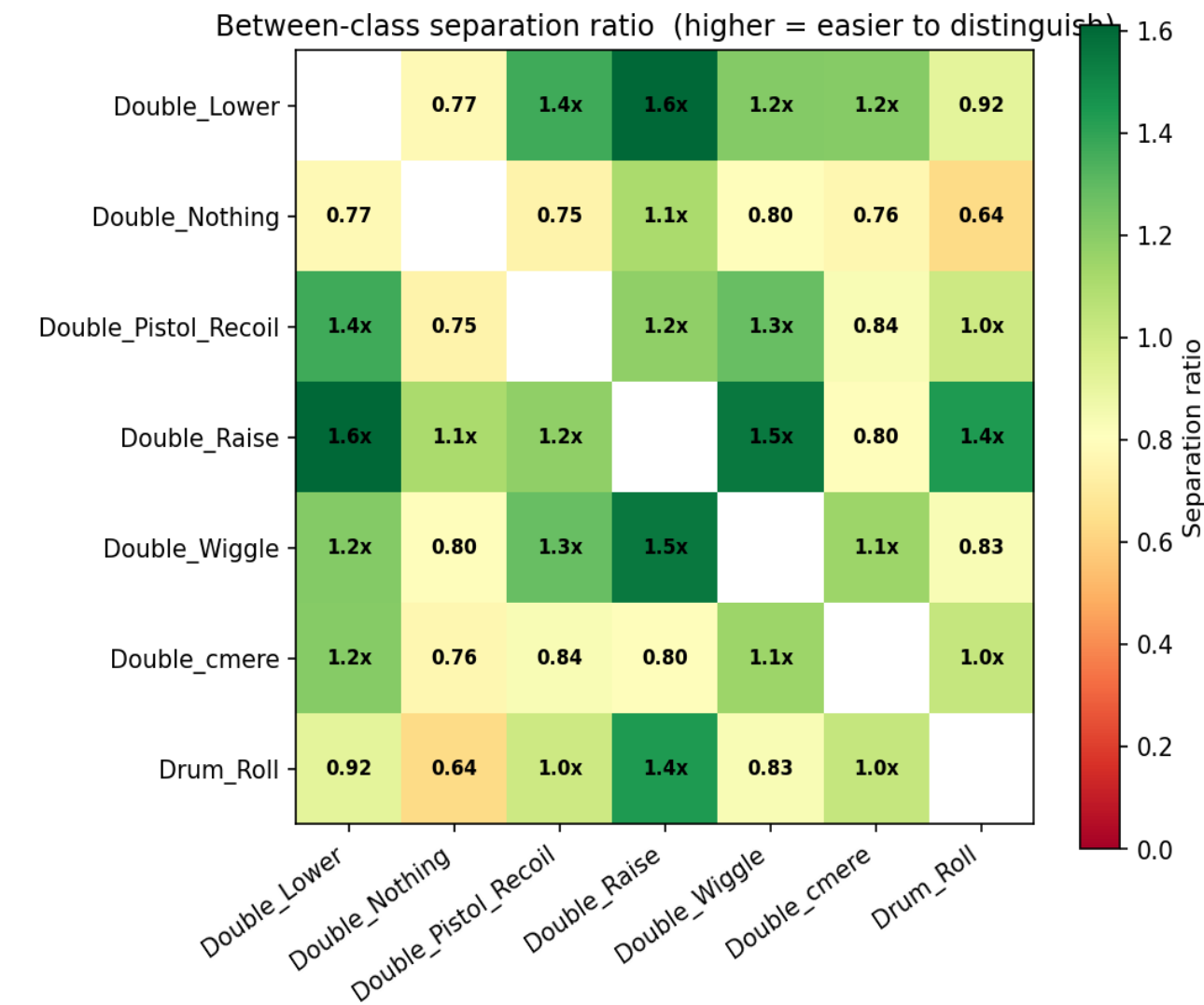
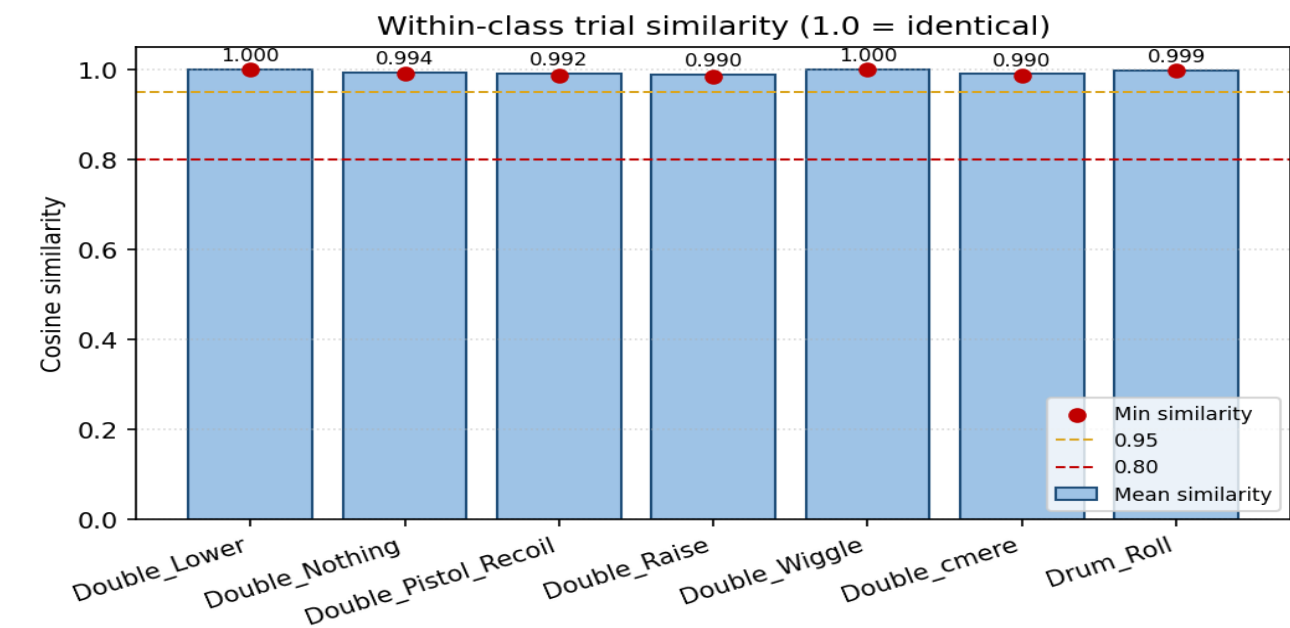


t-SNE feature space projection

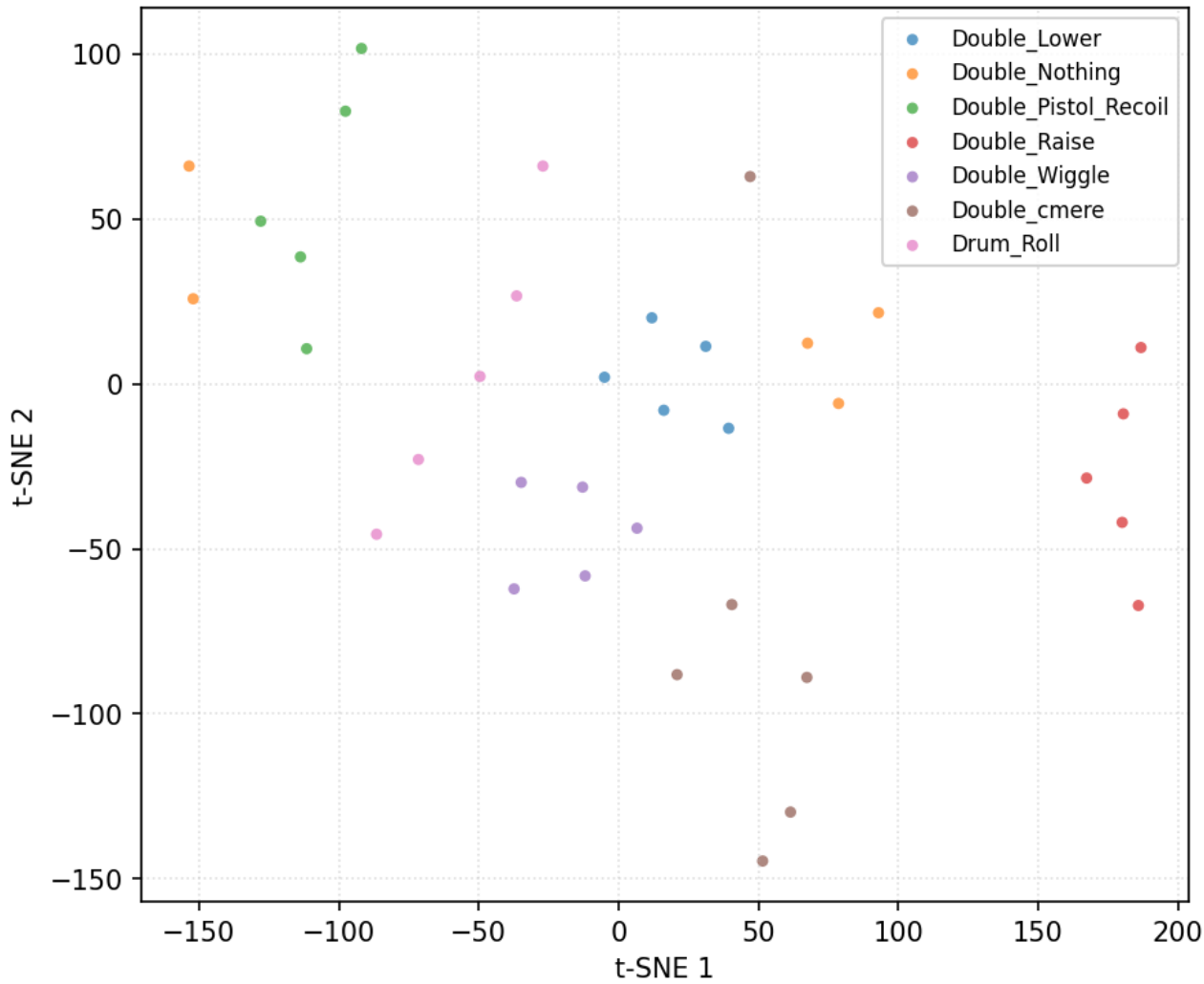


Data diagnostics — user 3_Anghad

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

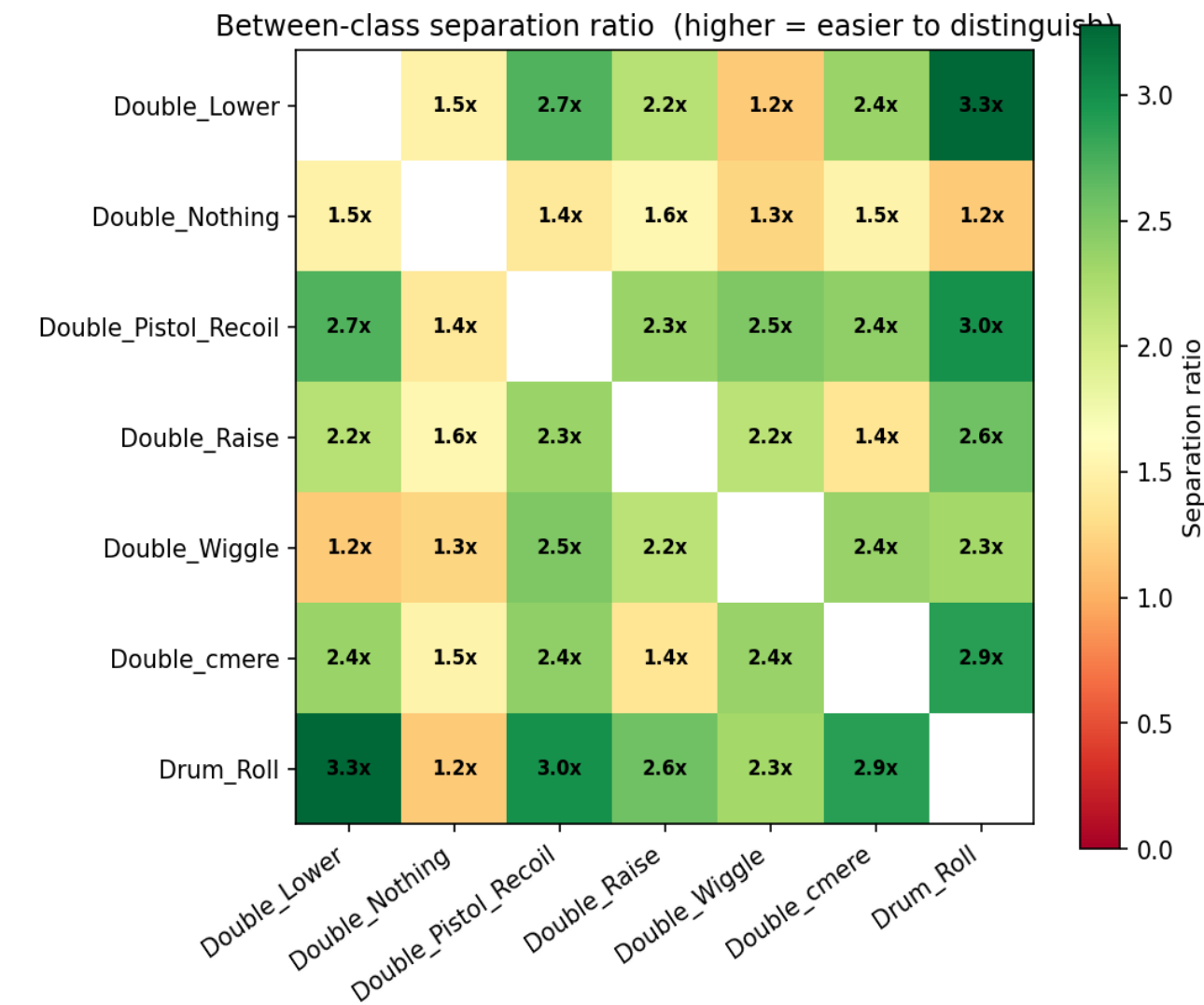
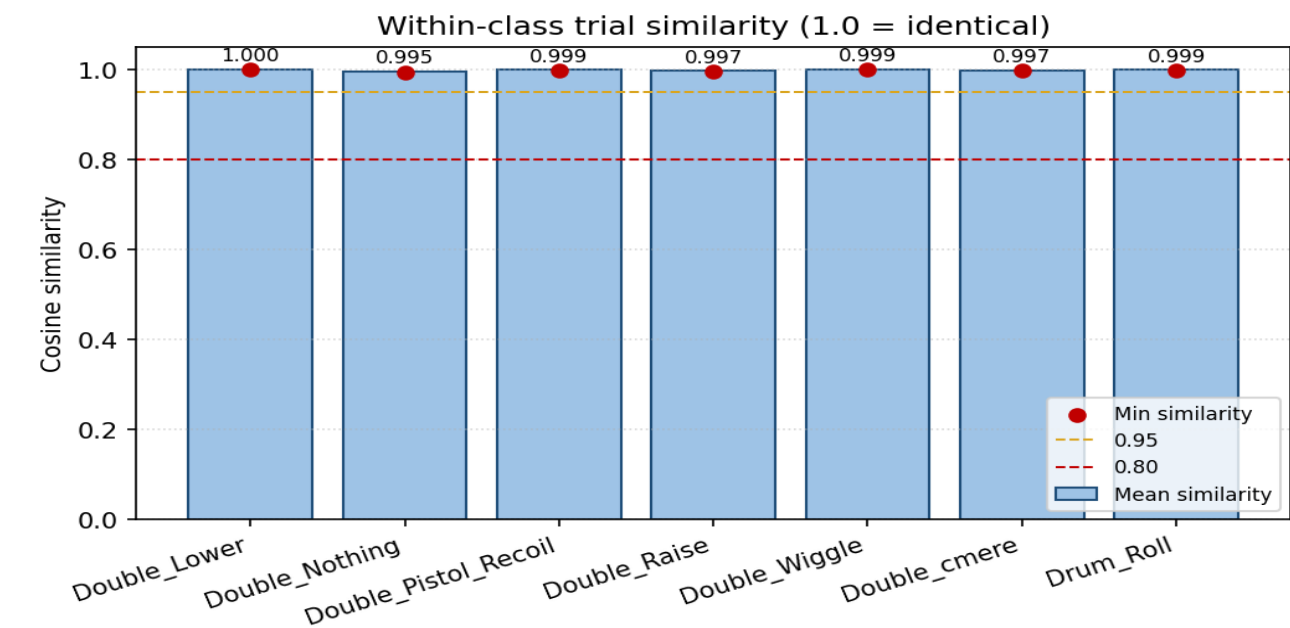


t-SNE feature space projection

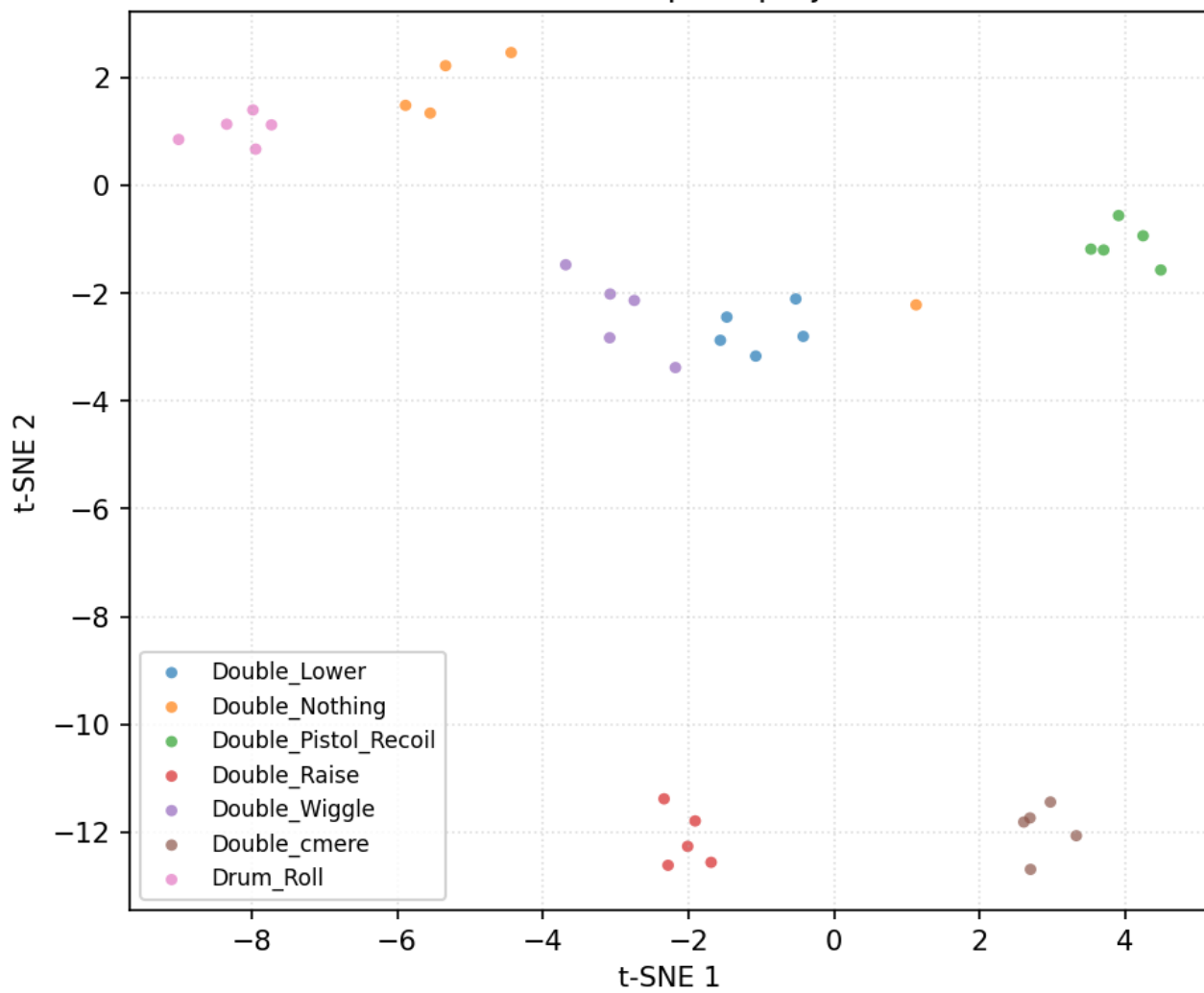


Data diagnostics — user 4_Daniel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

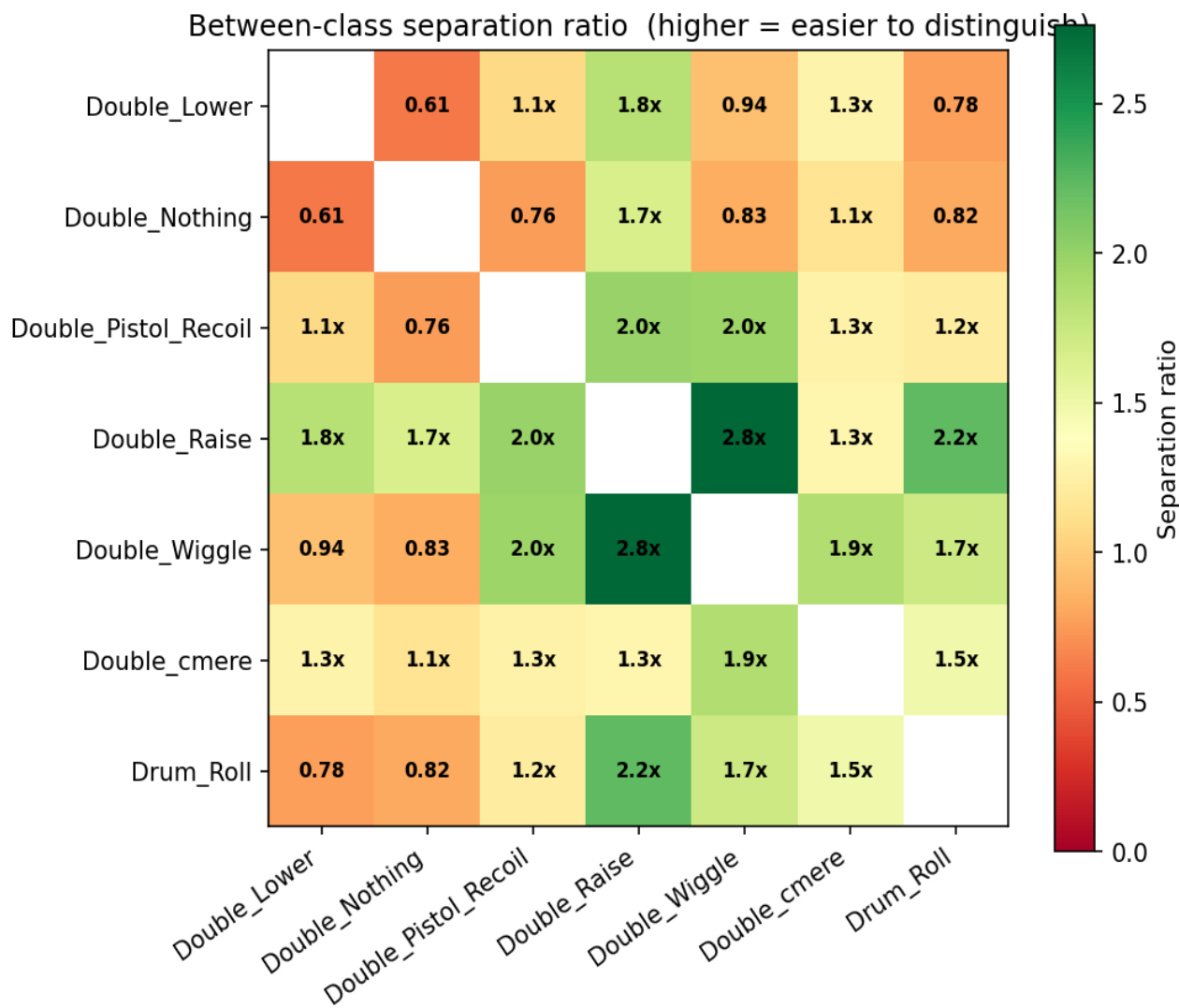
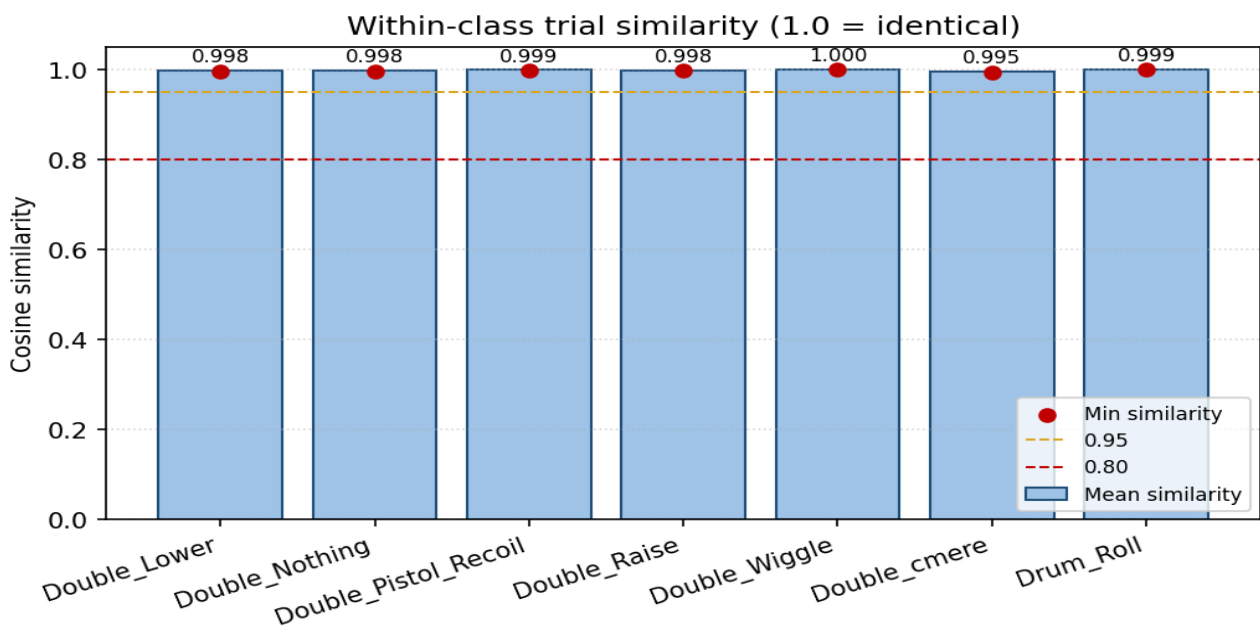


t-SNE feature space projection

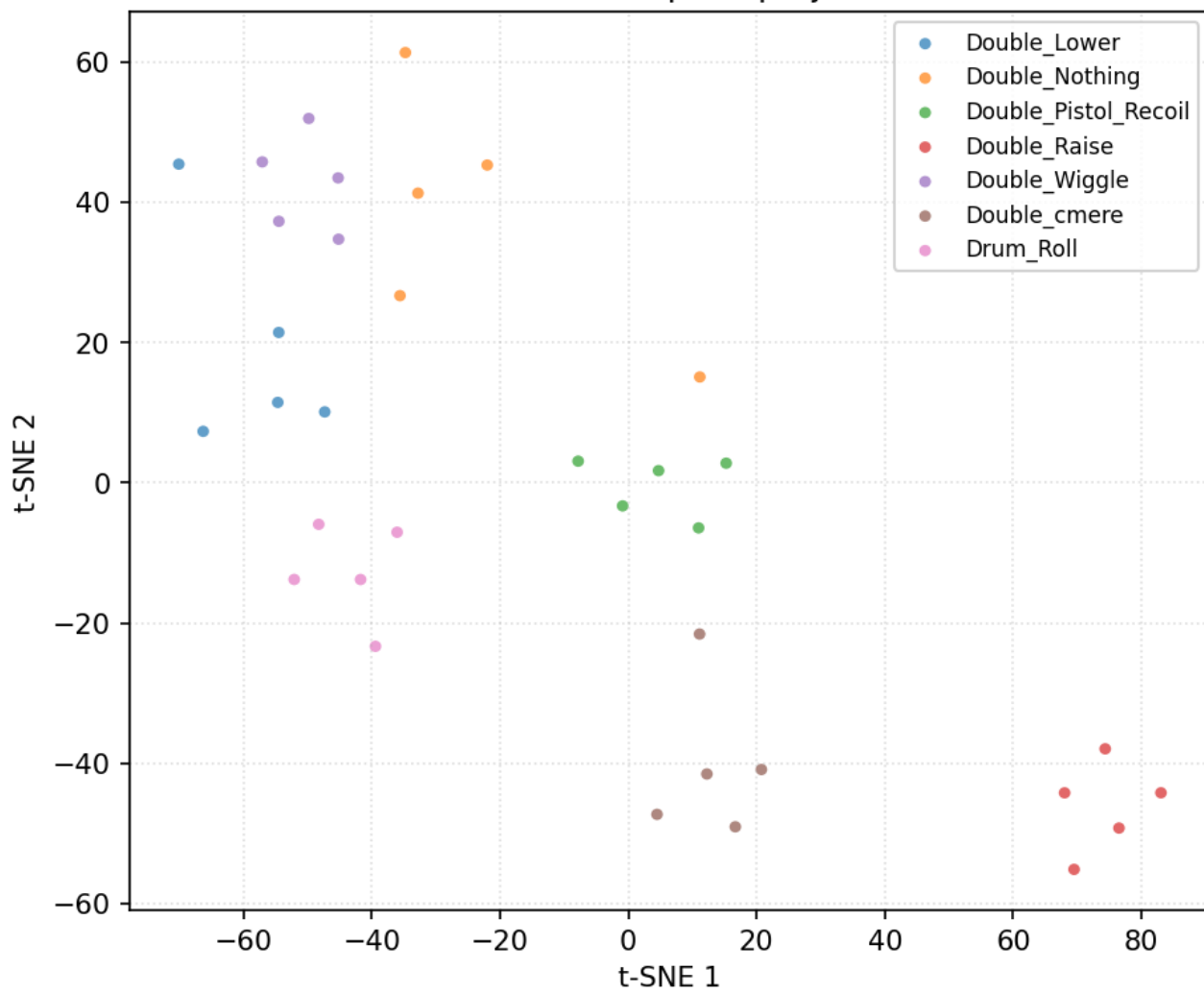


Data diagnostics — user 6_Harry

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

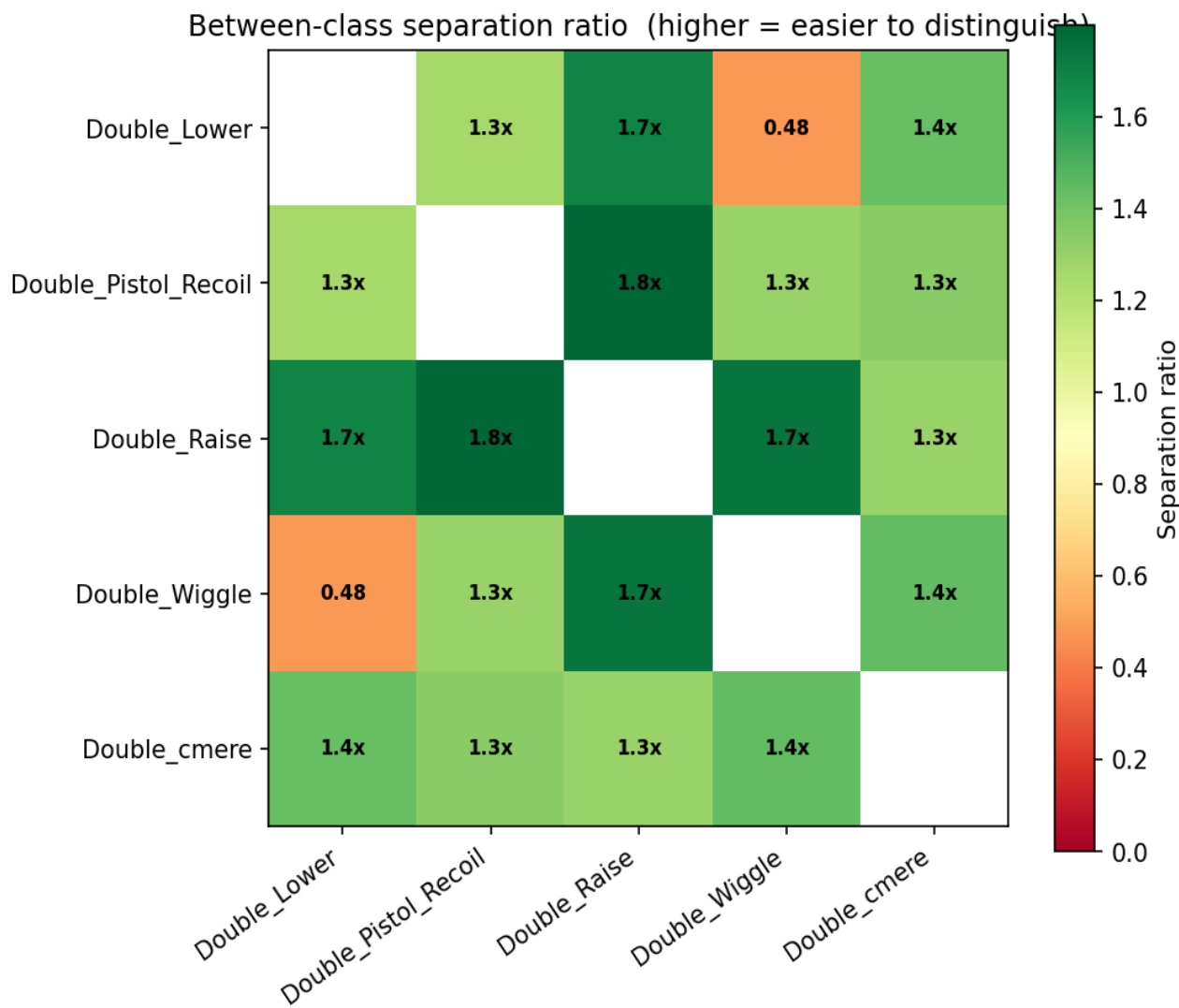
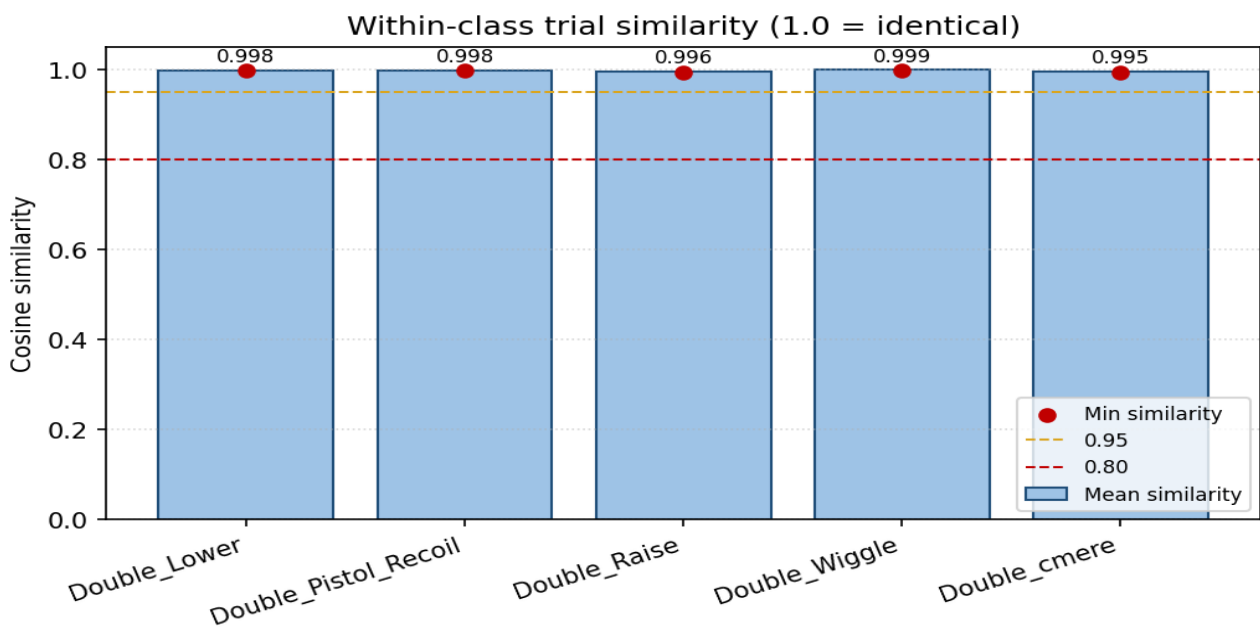


t-SNE feature space projection

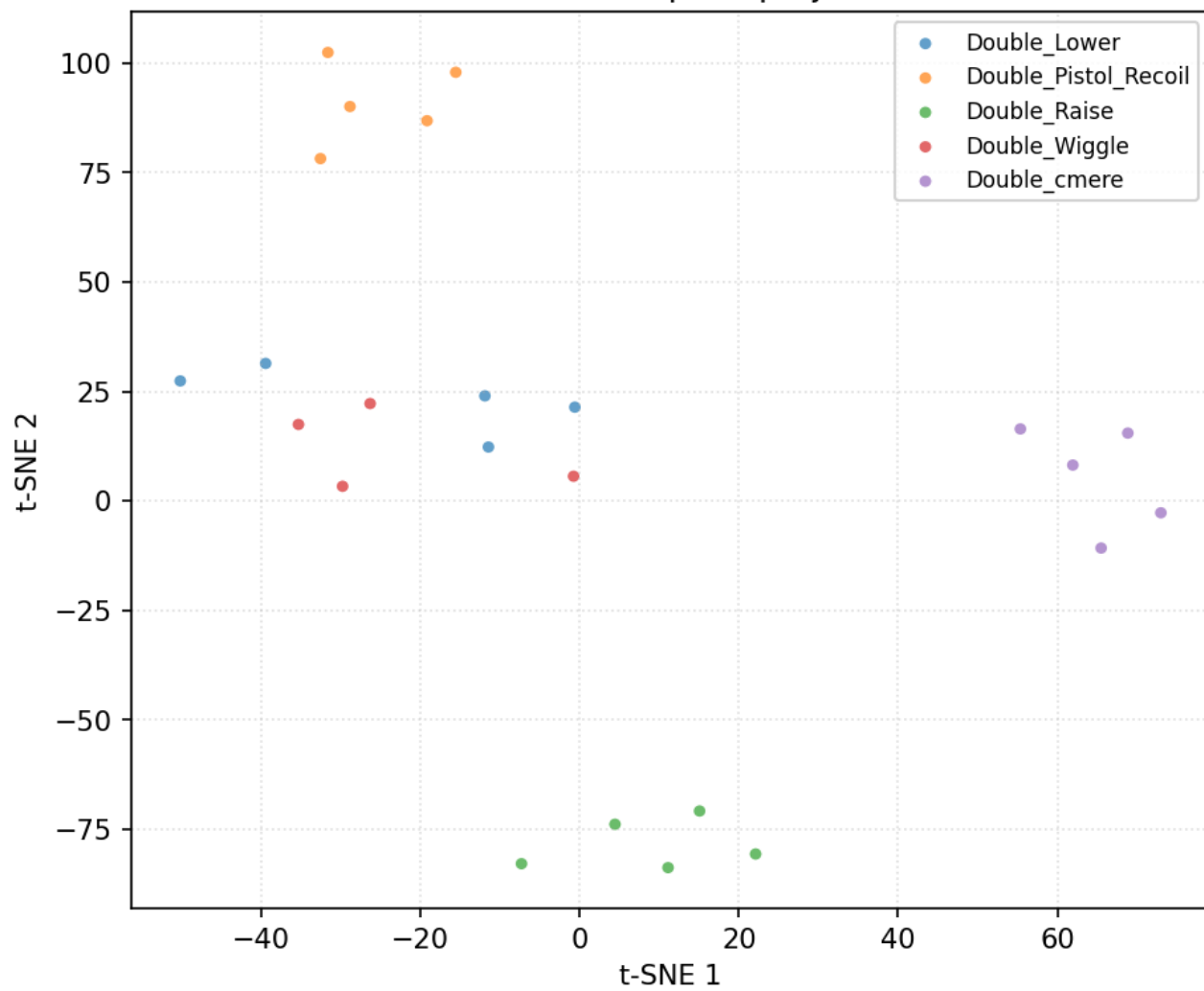


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
LOUO mean ± std	0.7355 ± 0.1565
Folds run	13
Inner-val mode	user
Per-trial norm	none

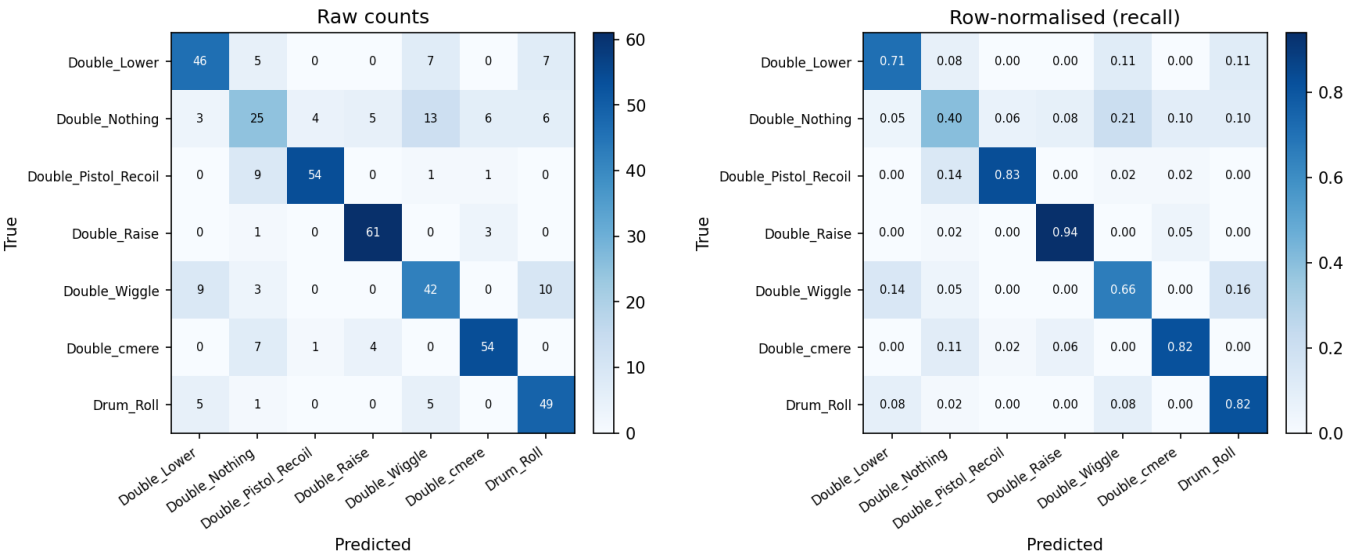
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.9143
2	12_Marcus	14_StephenV2	1128	35	0.7714
3	14_StephenV2	15_Tash	1128	36	0.6389
4	15_Tash	16_Bella	1128	35	0.7714
5	16_Bella	17_Raquel	1128	36	0.7500
6	17_Raquel	18_Bridgette	1131	35	0.6857
7	18_Bridgette	1_Alan	1131	35	0.6857
8	1_Alan	2_Alex	1131	35	0.4857
9	2_Alex	3_Anghad	1128	35	0.9143
10	3_Anghad	4_Daniel	1128	36	0.9167
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.5143
13	8_Jestin	11_Mansh	1164	24	0.5417

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.730	0.708	0.719	65
Double_Nothing	0.490	0.403	0.442	62
Double_Pistol_Recoil	0.915	0.831	0.871	65
Double_Raise	0.871	0.938	0.904	65
Double_Wiggle	0.618	0.656	0.636	64
Double_cmere	0.844	0.818	0.831	66
Drum_Roll	0.681	0.817	0.742	60
macro avg	0.736	0.739	0.735	447
weighted avg	0.738	0.740	0.737	447

Confusion matrix (aggregated across folds)



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
LOUO mean ± std	0.7206 ± 0.2161
Folds run	13
Inner-val mode	user
Per-trial norm	none

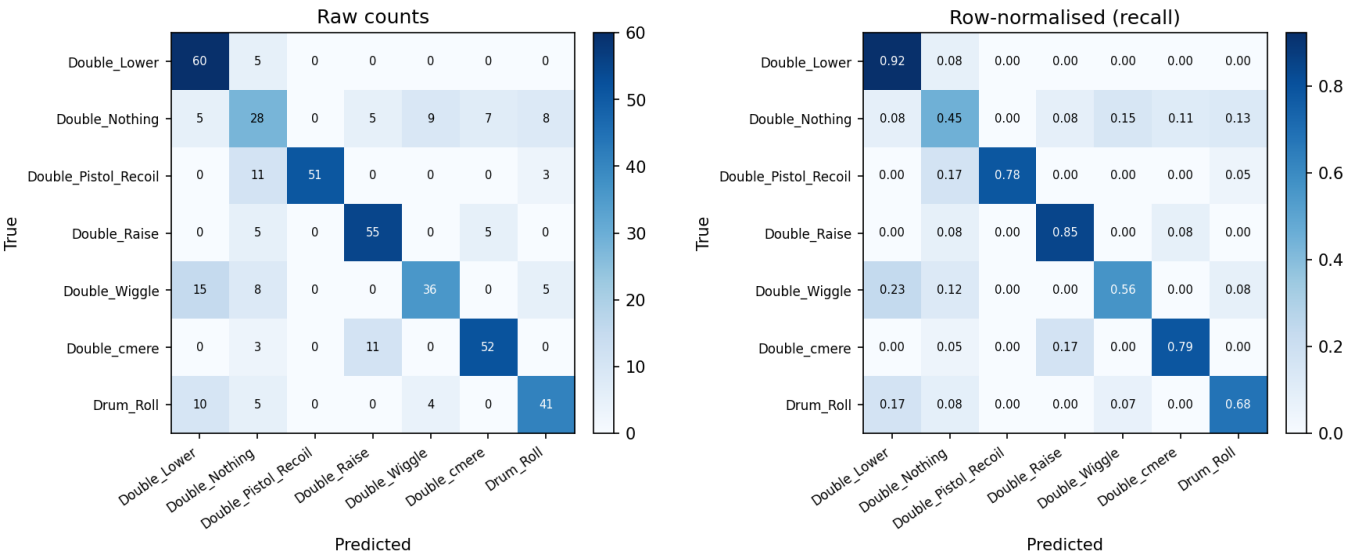
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.7714
2	12_Marcus	14_StephenV2	1128	35	0.7714
3	14_StephenV2	15_Tash	1128	36	0.7500
4	15_Tash	16_Bella	1128	35	0.4571
5	16_Bella	17_Raquel	1128	36	0.8333
6	17_Raquel	18_Bridgette	1131	35	1.0000
7	18_Bridgette	1_Alan	1131	35	0.3143
8	1_Alan	2_Alex	1131	35	0.3429
9	2_Alex	3_Anghad	1128	35	0.6857
10	3_Anghad	4_Daniel	1128	36	0.8889
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.9143
13	8_Jestin	11_Mansh	1164	24	0.6667

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.667	0.923	0.774	65
Double_Nothing	0.431	0.452	0.441	62
Double_Pistol_Recoil	1.000	0.785	0.879	65
Double_Raise	0.775	0.846	0.809	65
Double_Wiggle	0.735	0.562	0.637	64
Double_cmere	0.812	0.788	0.800	66
Drum_Roll	0.719	0.683	0.701	60
macro avg	0.734	0.720	0.720	447
weighted avg	0.736	0.723	0.723	447

Confusion matrix (aggregated across folds)



Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
LOUO mean ± std	0.7150 ± 0.2035
Folds run	13
Inner-val mode	user
Per-trial norm	none

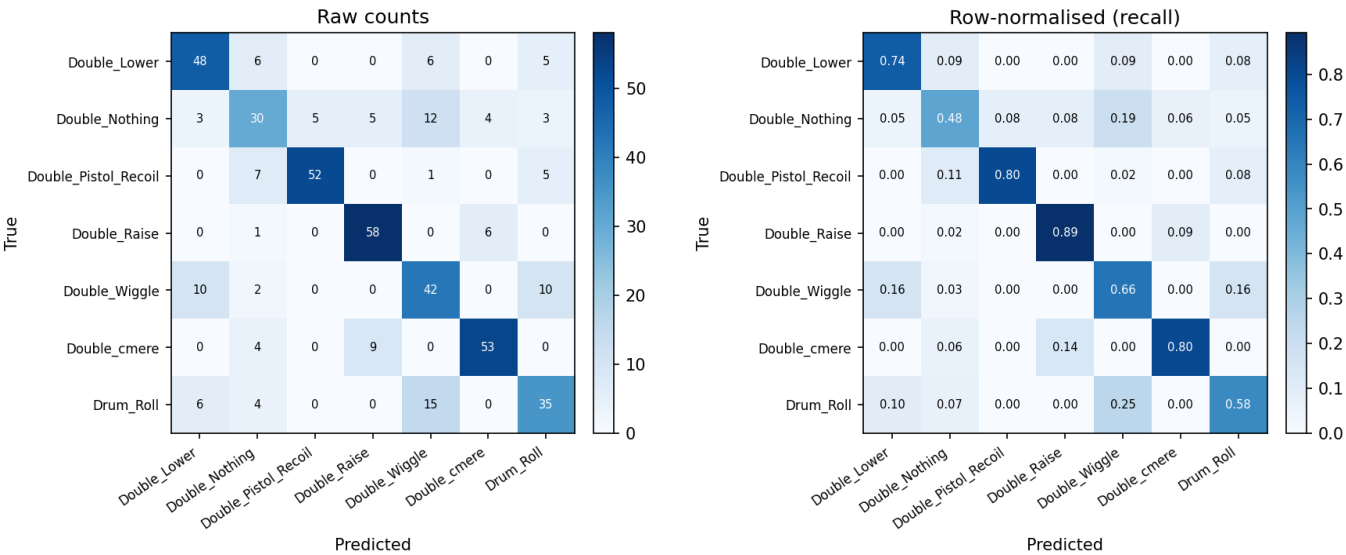
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.6000
2	12_Marcus	14_StephenV2	1128	35	0.5714
3	14_StephenV2	15_Tash	1128	36	0.6111
4	15_Tash	16_Bella	1128	35	0.6857
5	16_Bella	17_Raquel	1128	36	0.8056
6	17_Raquel	18_Bridgette	1131	35	0.7143
7	18_Bridgette	1_Alan	1131	35	0.5143
8	1_Alan	2_Alex	1131	35	0.2286
9	2_Alex	3_Anghad	1128	35	0.8857
10	3_Anghad	4_Daniel	1128	36	0.8611
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.9714
13	8_Jestin	11_Mansh	1164	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.716	0.738	0.727	65
Double_Nothing	0.556	0.484	0.517	62
Double_Pistol_Recoil	0.912	0.800	0.852	65
Double_Raise	0.806	0.892	0.847	65
Double_Wiggle	0.553	0.656	0.600	64
Double_cmere	0.841	0.803	0.822	66
Drum_Roll	0.603	0.583	0.593	60
macro avg	0.712	0.708	0.708	447
weighted avg	0.715	0.711	0.711	447

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
LOUO mean ± std	0.7050 ± 0.1587
Folds run	13
Inner-val mode	user
Per-trial norm	none

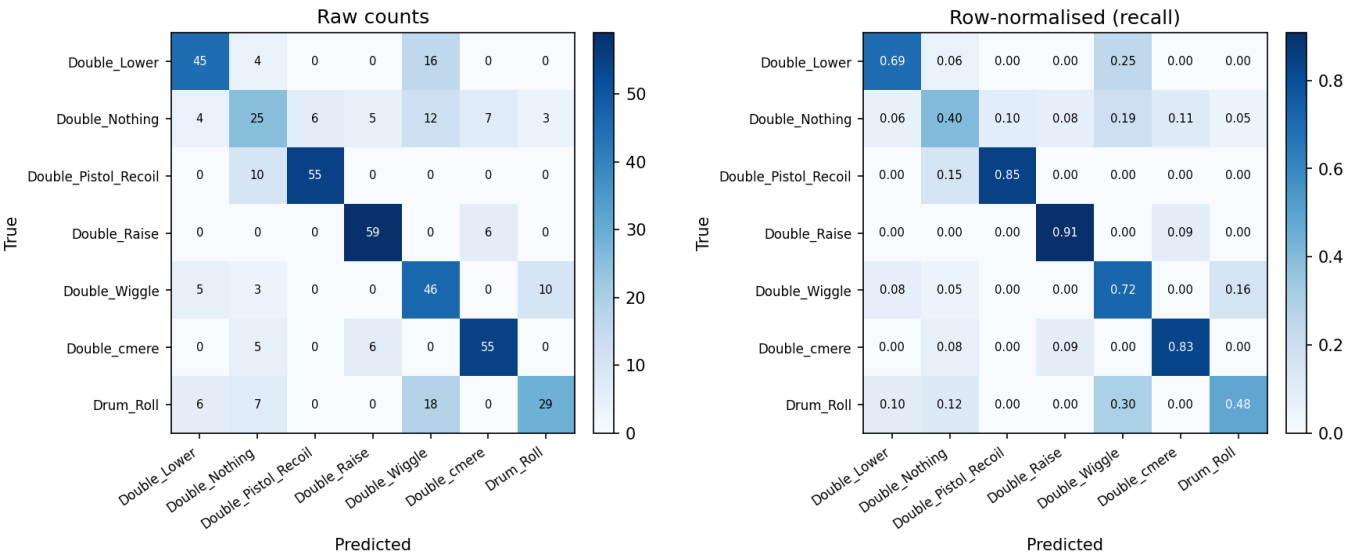
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.6857
2	12_Marcus	14_StephenV2	1128	35	0.6000
3	14_StephenV2	15_Tash	1128	36	0.8056
4	15_Tash	16_Bella	1128	35	0.4286
5	16_Bella	17_Raquel	1128	36	0.6944
6	17_Raquel	18_Bridgette	1131	35	0.6571
7	18_Bridgette	1_Alan	1131	35	0.6857
8	1_Alan	2_Alex	1131	35	0.5143
9	2_Alex	3_Anghad	1128	35	0.8857
10	3_Anghad	4_Daniel	1128	36	0.8889
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.5143
13	8_Jestin	11_Mansh	1164	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.750	0.692	0.720	65
Double_Nothing	0.463	0.403	0.431	62
Double_Pistol_Recoil	0.902	0.846	0.873	65
Double_Raise	0.843	0.908	0.874	65
Double_Wiggle	0.500	0.719	0.590	64
Double_cmere	0.809	0.833	0.821	66
Drum_Roll	0.690	0.483	0.569	60
macro avg	0.708	0.698	0.697	447
weighted avg	0.711	0.702	0.701	447

Confusion matrix (aggregated across folds)



Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
LOUO mean $\pm$ std	0.6629 $\pm$ 0.1500
Folds run	13
Inner-val mode	user
Per-trial norm	none

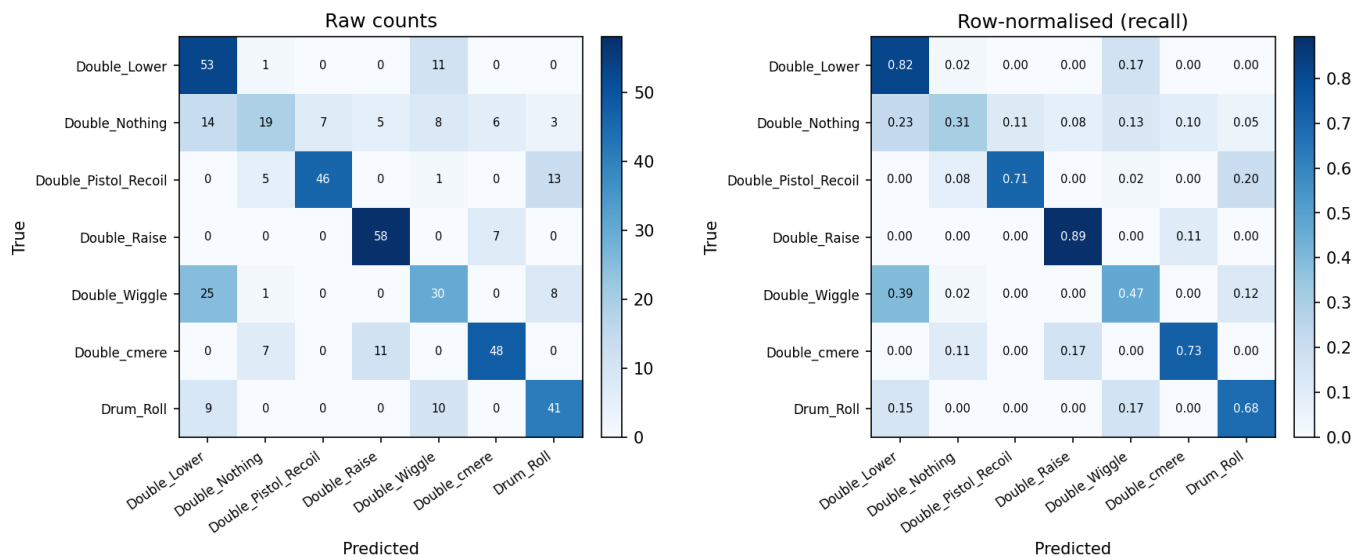
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.6571
2	12_Marcus	14_StephenV2	1128	35	0.5143
3	14_StephenV2	15_Tash	1128	36	0.6389
4	15_Tash	16_Bella	1128	35	0.5429
5	16_Bella	17_Raquel	1128	36	0.6944
6	17_Raquel	18_Bridgette	1131	35	0.8000
7	18_Bridgette	1_Alan	1131	35	0.6571
8	1_Alan	2_Alex	1131	35	0.4286
9	2_Alex	3_Anghad	1128	35	0.7429
10	3_Anghad	4_Daniel	1128	36	0.7500
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.4286
13	8_Jestin	11_Mansh	1164	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.525	0.815	0.639	65
Double_Nothing	0.576	0.306	0.400	62
Double_Pistol_Recoil	0.868	0.708	0.780	65
Double_Raise	0.784	0.892	0.835	65
Double_Wiggle	0.500	0.469	0.484	64
Double_cmere	0.787	0.727	0.756	66
Drum_Roll	0.631	0.683	0.656	60
macro avg	0.667	0.657	0.650	447
weighted avg	0.669	0.660	0.652	447

Confusion matrix (aggregated across folds)



Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
LOUO mean ± std	0.6441 ± 0.1621
Folds run	13
Inner-val mode	user
Per-trial norm	none

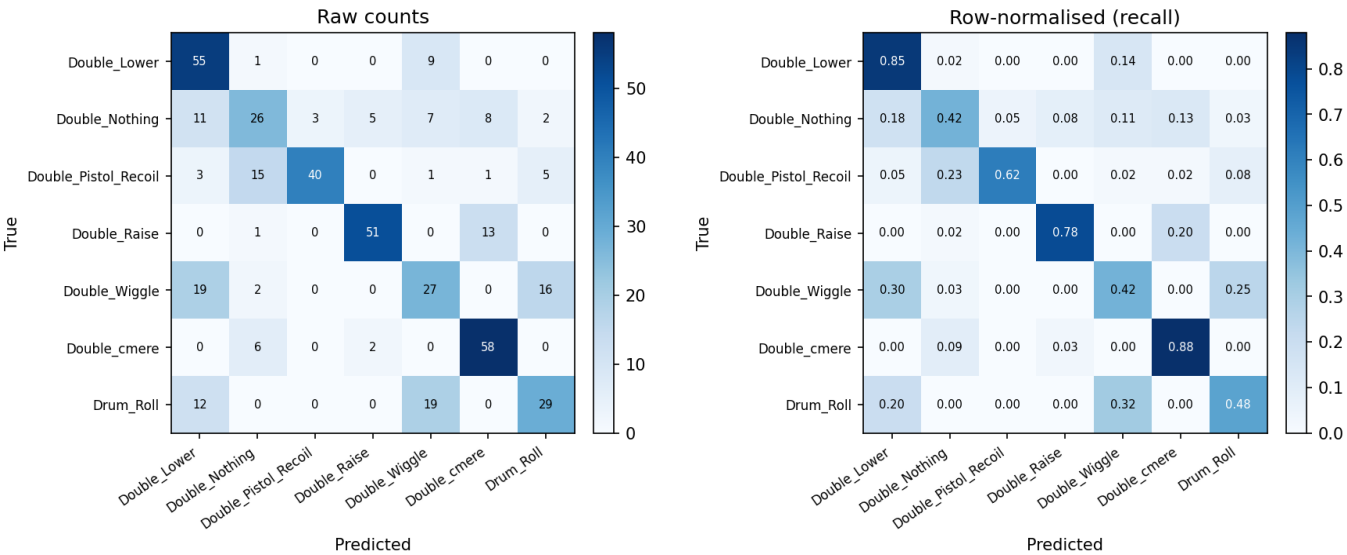
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.5429
2	12_Marcus	14_StephenV2	1128	35	0.7429
3	14_StephenV2	15_Tash	1128	36	0.6944
4	15_Tash	16_Bella	1128	35	0.5143
5	16_Bella	17_Raquel	1128	36	0.6944
6	17_Raquel	18_Bridgette	1131	35	0.7429
7	18_Bridgette	1_Alan	1131	35	0.4571
8	1_Alan	2_Alex	1131	35	0.3714
9	2_Alex	3_Anghad	1128	35	0.6571
10	3_Anghad	4_Daniel	1128	36	0.6944
11	4_Daniel	6_Harry	1131	35	0.9714
12	6_Harry	8_Jestin	1164	35	0.4571
13	8_Jestin	11_Mansh	1164	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.550	0.846	0.667	65
Double_Nothing	0.510	0.419	0.460	62
Double_Pistol_Recoil	0.930	0.615	0.741	65
Double_Raise	0.879	0.785	0.829	65
Double_Wiggle	0.429	0.422	0.425	64
Double_cmere	0.725	0.879	0.795	66
Drum_Roll	0.558	0.483	0.518	60
macro avg	0.654	0.636	0.633	447
weighted avg	0.657	0.640	0.637	447

Confusion matrix (aggregated across folds)



Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
LOUO mean ± std	0.6431 ± 0.1689
Folds run	13
Inner-val mode	user
Per-trial norm	none

LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	1131	35	0.6571
2	12_Marcus	14_StephenV2	1128	35	0.5714
3	14_StephenV2	15_Tash	1128	36	0.6667
4	15_Tash	16_Bella	1128	35	0.5143
5	16_Bella	17_Raquel	1128	36	0.5278
6	17_Raquel	18_Bridgette	1131	35	0.6857
7	18_Bridgette	1_Alan	1131	35	0.5429
8	1_Alan	2_Alex	1131	35	0.2571
9	2_Alex	3_Anghad	1128	35	0.6286
10	3_Anghad	4_Daniel	1128	36	0.8889
11	4_Daniel	6_Harry	1131	35	0.9429
12	6_Harry	8_Jestin	1164	35	0.6857
13	8_Jestin	11_Mansh	1164	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.574	0.600	0.586	65
Double_Nothing	0.489	0.355	0.411	62
Double_Pistol_Recoil	0.847	0.769	0.806	65
Double_Raise	0.789	0.692	0.738	65
Double_Wiggle	0.452	0.656	0.535	64
Double_cmere	0.736	0.803	0.768	66
Drum_Roll	0.660	0.583	0.619	60
macro avg	0.650	0.637	0.638	447
weighted avg	0.651	0.640	0.640	447

Confusion matrix (aggregated across folds)

